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HUGO GERNSBACK
Editor

AIR STORIES

WONDER



Science-Aviation Stories
by

ED EARL REPP
EDMOND HAMILTON
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Cities in the Air

By Edmond Hamilton

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Edmond Hamilton

Here is one of the most extraordinary stories that it has been our good fortune to read. For sheer audacity in construction, excellence in science and breath-taking adventure, this story undoubtedly stands in the foreground of science air-fiction stories of the year.

The recent advances in aeronautics where airplanes have been in the air for weeks at a time without coming down to the ground, point the way for tremendous achievements in the generations to come.

City life today is a conglomeration of structures close together. We have buildings now that house as many as 40,000 people at one time and soon we will have single business buildings that will house 100,000 and more individuals at the same time. Furthermore, every doctor will tell you that living at the surface of the earth is usually unhealthy because of the dust and the high density of the air, which gives rise to most pulmonary diseases, particularly consumption, colds and the like. At high altitudes such diseases tend to disappear. Therefore physicians usually send their afflicted patients to the higher altitudes.

You may be sure that conditions such as are described by the author of this marvelous story will come about sooner or later.

We also know that this story will arouse a great storm of discussion among our readers, due particularly to the audacity of the author in picturing his

ideas as to future aviation—which by the way will not seem so fantastic two hundred years hence as they might seem now.

"Captain Martin Brant, of American Federation Air-Cruiser 3885!"

As the high clear voice rang through the bridge-room of my racing cruiser, I turned toward the distance-phone from which it issued. Pressing a stud beneath the instrument I answered into it.

"Captain Brant speaking."

"Order of the First Air Chief to Captain Brant: You are informed that the European and Asiatic Federations have combined in alliance to launch a great and unexpected attack upon the American Federation. The European Federation fleet of five thousand air-cruisers is now racing over the Atlantic toward New York and other eastern cities, while the Asiatic Federation fleet of the same size is heading over the Pacific toward our western coasts. All American cruisers patrolling east of the Mississippi, including your own, are ordered to head at full speed toward New York, where our eastern squadrons are assembling to meet the European Federation fleet. Upon arriving there yourself and all other squadron commanders will report at once to the First Air Chief."

The clear voice ceased, and I turned from the distance-phone to meet the startled eyes of Macklin, my first officer, who stood at the cruiser's wheel beside me.

"Head eastward—full speed, Macklin!" I cried to him. "It's war at last—war with the European and Asiatic Federations!"

Instantly Macklin swung over the wheel in his hands, and as he did so the whole long bulk of our cruiser swung likewise in mid-air, curving up and backward to race eastward above the green plains, the descending sun at our backs. A moment more and the cruiser's long torpedo shape, gleaming and unbroken metal save for the rows of portholes and the raised, transparent-walled bridge-room in which we stood, was splitting the air eastward at a speed that mounted with each moment. I reached for the order-phone, and as Hilliard, my young second officer, answered from the motor-rooms

beneath, I informed him briefly of what had just been told me. Then there was a muffled cheer from the hundred-odd members of our crew, beneath, and a few minutes later the drone of the great motors had reached to an even higher pitch, and we were racing through the sunlight high above the earth at more than a thousand miles an hour.

Standing there with Macklin in the bridge-room as we shot eastward, though, my thoughts were grave enough despite the exciting quality of the news we had just heard. War!—the war that we of the American Federation had expected, had feared for decades. It had not been more than thirty years since the third Air War of 2039. Three mighty nations alone now shared the world between them; the American Federation, comprising the whole North and South American continents, with New York as its capital; the European Federation, which included all Europe west of Caucasus and all Africa, its center at Berlin; and the Asiatic Federation, which held all Asia and Australasia for the brown and yellow races, with Peking as its capital.

And though for three decades now there had been peace between them, it had been an uneasy peace dictated by the fact that each feared to attack another lest he be attacked by the third. The great navies of air-cruisers of the three mighty Federations had patrolled the air in ceaseless vigilance, their air-forts ever watchful. Lately, however, it had become apparent to all that a rapprochement had taken place between the European and Asiatic Federations, and such an alliance could only mean an attack upon our own, the American. So we had stood even more vigilantly upon the watch, and now that for which we had waited had come at last, and the two great Federations had launched their two mighty fleets upon us.

Gazing ahead, as our cruiser drove onward, I was as silent as Macklin, at the wheel beside me, and as young Hilliard, who had come up into the bridge-room from beneath. Far beneath us the green plains were rolling swiftly backward, as our motors hummed their unceasing song of power. Those great electric motors drew their current in limitless quantities from the electrostatic or atmospheric electricity surrounding the earth, by means of great transformers that changed it from electrostatic to current electricity to give us a power that could hurl us forward with almost unlimited endurance and speed. Connected as they were to our great horizontal tube-propellers, which were set in the cruiser's walls and which moved it forward by drawing immense volumes of air at vast speed through themselves from

ahead, those motors could fling us on at more than a thousand miles an hour. This utmost force, as our indicators told us, was shooting us eastward now.

Beneath us the green plains had given way to the great tumbled folds and peaks of the Alleghanies. Somewhere to the south lay Pittsburgh, and to the north Cleveland and Buffalo, but being headed directly to New York, we therefore did not see them. Beneath us we could make out in swift flashes of vision masses of the air-traffic between those cities, great passenger-liners and bulky freight-carriers and slender private craft, but in our own military-craft level there moved only a few cruisers like our own racing eastward toward New York in answer to the alarm. With these, however, there was small danger of collision.

Now the Alleghanies had dropped behind and we were rocketing over the rolling, pleasant countryside that lies between them and the eastern Appalachians. As we shot on I gazed downward, over the green and silent and empty landscape rushing beneath us, and wondered momentarily what a citizen of fifty years ago would have thought to see this once-populous land over which we were speeding lying as lifeless and deserted beneath us as it was now. Then it had given way to the greater folds and ridges of the Appalachians, and then, as we shot on and over their tumbled masses, Macklin lifted his hand from the wheel to point ahead.

"The air-forts!" he said.

On to New York

Swiftly they were looming before us as we rushed on toward them, giant domed cubes of dull metal, each five hundred feet in width, that hung in a great, curving line in mid-air before us, five miles above the green land. At intervals of five miles they hung, floating motionless there in a great grim chain or ring, the metal sides and dome of each bristling with great heat-guns like those of our own cruisers, and with narrow openings from which the occupants could gaze forth. Each of these great air-forts, we knew, was suspended thus high above the ground by the gravity-repelling effect of the

cosmic rays. It had been but fifty years since the machinery had been discovered which directed the great power of the cosmic rays to overcome the force of gravity. It had been found that the rays could be collected and their power concentrated in the structures that they were to support. Dynamic towers were used for the collection of this great limitless energy.

In a great ring they hung before us, the line of them curving away vastly to right and left, a great ring that encircled and defended New York, as air-forts hang in rings about all air-cities for defence. Then as we drove toward the nearest of these great fortresses of the air, there came from the distance-phone before me its sharp challenge.

Swiftly I replied to that challenge and then we were driving past the air-fort, past the openings in its walls through which we could see those inside standing ready at the great heat-guns. We heard faintly their cheers as we flashed past them toward the east; we cheered ourselves somewhat by the sight of the air-forts. They could maneuver in space in any direction, though at only a fraction of the speed of the air-cruisers. They would form a stubborn defence for New York, we knew, though incapable of meeting alone a swift invading fleet. But now far ahead, as we rushed within the mighty ring of the air-forts, we glimpsed the gray gleam of the Atlantic's vast expanse, stretching away to the east, the green, irregular coastline, the narrow little island, between a larger island and that coast, that had been the site of the New York of fifty years before. Green and deserted as all the countryside behind us it lay now, but I glanced at it only, looking up as there came a low exclamation from Hilliard, beside me.

"New York!"

Full before us lay the mighty city, now, waxing with each moment greater as we raced on toward it. The air about and beneath us was filled with the great swarms of cruisers like our own and of merchant-traffic that was converging from north and west and south upon it. For the moment we three gazed toward it, forgetful of the peril that had brought us to it. We were caught and entranced as always by the splendid and superb beauty of this New York. For it was a New York immeasurably different from that city upon the earth, that decades ago had born its name. It was a city, not of the earth, but of the air.

It was a city whose close-clustered spires and towers and pyramids had been gathered together upon a vast metal disk-like base, and hung suspended five miles above the green earth! It was circular in form and of five miles diameter, the colossal metal base or disk upon which it rested more than a thousand feet in thickness, the metal buildings and towers that rose from that base and were integral with it soaring for five thousand feet farther upward! A colossal city floating there in the air, with its streets and buildings swarming with activity, with thronging hordes, and with great masses of fear-driven craft speeding through the air toward it from all directions.

A city of the air! Suspended by huge batteries of great electrostatic motors in its base, motors that drew the exhaustless energy of earth's atmospheric electricity from countless slender pinnacles that soared from the central plaza; whence the current was conducted along cables within the pinnacles to the giant motors beneath. The cities too were suspended by the gravity-repelling quality of the collected cosmic rays. To this had mankind come, at last. The flimsy airplanes of a good century before, with their little endurance records of weeks and months in the air, had given way to the great electric-driven cruisers which drew their power from the static about them, and which could stay aloft indefinitely. And then had come the great air-forts, held aloft in the same way, and finally, when the great air-wars had made life upon the ground so unsafe as to approach suicide, then had come the construction of giant metal cities, on huge metal bases, that contained enough great motors and tube-propellers to hold themselves in any direction at moderate speed.

The Great Conference

Such now were all the cities of earth; Chicago, San Francisco, Buenos Aires, Berlin and Tokio. Great cities that hung always in mid-air, usually near the sites of those vanished cities of earth from which they had gained their names. But the air-cities could move from place to place now and then for better climate or defence. These great cities held within them all the world's population; the earth beneath being used only for the mining of

metallic ores and the minerals used in the creation of the synthetic foods and fabrics now universal.

The great cities were each protected by a ring of air-forts, and also by great batteries of heat-guns set within their own walls. A hundred of such huge air-cities there were in the American Federation, holding in their colossal masses of clustered sky-flung towers an average of five million inhabitants each. And in the European and Asiatic Federations combined, I knew, there were more than two hundred mighty cities of the air.

Now, though, it was the huge air-city of New York that held all our attention, and as we rushed closer toward it I saw that above it, above the panic-driven masses of air-craft that were swirling down to take refuge within it, there hung squadron upon squadron of cruisers like our own, over two thousand in number, hanging there in grim, motionless ranks, as though unconscious of the swarming, fear-driven activity in the huge city beneath them. From every quarter other cruisers were arriving to join those squadrons, cruisers that came like our own from patrols over the green inland plains, from above the icy Labrador wastes, from over the jungle-bordered Caribbean coasts, all rushing to answer the call to arms. As our own ship neared the city, we headed down toward the central plaza.

"Straight down to the central plaza, Macklin," I said. "The First Air Chief will be there and orders are to report to him first."

Macklin had already slowed our ship's speed, and now as we drove to a position beside the aspiring central pinnacle, with its clustered points, the city's static-tower, he turned the power of our motors completely from our horizontal tube-propellers, into our vertical ones, which held us motionless in mid-air. Then, as he slowly decreased that power, we sank smoothly down until in a moment more we had come to rest upon the smooth central plaza among a score or more of other cruisers. These rested in a great ring about the plaza's edge, their crews waiting within them, but at the center of that ring, beside the mighty static-tower's base, stood a little group of men, the First Air Chief, Yarnall, and his squadron-commanders.

As our cruiser came to rest I opened the door beneath the bridge-room, and stepped onto the metal plaza and across it toward that group. Around the great plaza, I noted, were vast, seething crowds, thousands upon thousands of the mighty air-city's inhabitants. Other thousands were gazing down

toward us from the towers that soared around us into the golden afternoon sunlight. These people, watching us and the mighty fleet hanging grimly far above, were silent, but from beyond them there came to my ears from far across the air-city's mighty mass, the dull roar of millions of blended voices, in unceasing, excited shouts. Then I reached the First Air Chief and the group before him, my hand snapped to a salute, which Yarnall silently returned. And then, gazing for a moment in silence from one to another of us, his strong face and gray eyes grave, he began to speak to us.

"You, the squadron-commanders of our eastern forces," he said, "know why you have been summoned here, why I, under the orders of the Federation's Central Council, have summoned here you and all the cruisers that wait above us. The great European Federation fleet, twice as large as our forces, is rushing westward over the Atlantic toward us, and within the hour we must meet that fleet in battle."

He paused, and in the silence that ensued the dull, dim roar of the great city about us seemed suddenly infinitely remote from our ears. Then the First Air Chief went on.

"Within the hour we must meet that fleet in battle and as we go out to meet it our western forces will be going out from San Francisco, under the command of the Second Air Chief, to meet the Asiatic Federation fleet racing eastward toward it. And upon those two battles rests now the fate of our nation. If they are lost, if either of them is lost, within days our nation will be but a memory, our cities annihilated. If the two approaching fleets are defeated and beaten back, then we shall have won for ourselves a respite in which we can prepare to meet the great enemies that crowd now upon us. So I say to you, the Central Council says to you, that this battle must not be lost!

"The fleet that we must meet has twice the number of cruisers of our own, and there have been rumors of some new method being prepared by them with which to attack us, now or later. We have to aid us only the air-forts about this city, which have been equipped with a new device. I have ordered them to move east of the city to lie between it and the enemy. This great air-city itself, when we go out from it, will move inland at its highest speed away from the battle, just as Boston and Charleston and Miami and San Francisco and Los Angeles and all our great air-cities, north and south.

There will be, therefore, none but our cruisers gathered above and our air-forts massing eastward to fight this battle upon which the Central Council has staked our fate.

"But great as these odds are against us, this battle must not be lost! We are the sons of the Americans who fought through the First and Second and Third Air Wars, who reared this nation out of the blood of a thousand air battles until now its hundred air-cities hold in their power a third of all the world. And now that the rest of that world comes against us, the Last Air War begins. My word to you is this: Fight only as those men before you fought, and before tomorrow the European Federation fleet shall have been beaten back—or the last of our cruisers and our air-forts and ourselves will have perished!"

There was silence as the First Air Chief ceased, and then from us assembled commanders there broke a great cheer, a cheer that was taken up by the massed thousands around the plaza and that spread like fire over all the great air-city about us. Then we all returned toward our waiting cruisers, the First Air Chief toward his own, and a moment later his cruiser, with its three parallel stripes of silver running from stem to stern distinguishing it from all others, was rising smoothly upward, followed by our own. Upward we shot, a vast roar coming up to us from the mighty floating city beneath us. Then our score or more of ships were taking their places each at the head of its squadron of a hundred ships, while the First Air Chief in his silver-striped flagship rushed to a position at the head of all. There we hung, the dull, great roar coming unceasingly up to us from the city below, and then as an order sounded from the distance—phones of all the fleet we were moving forward, eastward, out from over the great air-city, from over the green coastline, out over the gray expanse of the Atlantic.

With Macklin and Hilliard again beside me as our own cruiser moved forward at its squadron's head, we three turned to glance back. We saw New York, its mighty towers splendid against the descending sun, moving also, but slowly westward and away from us, away from the coming battle, dwindling to a dark spot and vanishing as we raced on outward over the gray Atlantic. Now we were racing above the great air-forts that had massed in a great double line a score of miles out from the coast, high above the waters. Over these too we sped, at steadily mounting speed, until with great motors droning, crews shouting as they ran our heat-guns out from tops and

sides and keels, winds whining shrill about us, our great fleet reached its maximum speed toward that great oncoming battle by which our Federation was to stand or fall.

CHAPTER II

The Battle Over the Atlantic

Gazing ahead, Macklin and Hilliard and I stood together in the bridge-room of our cruiser. The squadron which we headed was at the lead of one of our fleet's great columns. Far behind us stretched its ships, flashing forward at uniform speed. Then from the distance-phone before us came the First Air Chief's voice.

"Squadrons 1 to 6 take up scouting positions!" he ordered.

Instantly the first six squadrons of the two columns, our own one of the first, leapt forward and out from the two great lines of the main fleet. Our own and another squadron moved straight ahead, past the silver-striped flagship of the First Air Chief, until our two hundred ships had spread out into a great, thin fringe that was flying forward miles before the main body of our fleet. Two of the other four squadrons drove to right and left of the fleet, spreading there in the same way, the remaining two taking up positions high above and far beneath our two great columns. Thus, with its great lines of scouts fringing it and protecting it from surprise on all sides, our great fleet drove on toward the east over the gray and endless plain of the Atlantic, holding at Yarnall's orders a speed of eight hundred miles an hour.

The crimson descending sun flaming in the heavens behind us, the great gray ocean stretching endlessly beneath us, we rushed on through empty sea and sky. By then Hilliard had gone down to take up his position with the crew beneath, but Macklin and I still stared into the great empty vista before us. With the drone of our great motors and those of the scouts flying beside us, we seemed like a great flight of bees. Beneath there was no sound now from the crew, a silence that told of the tenseless, of expectancy. But still before us was no sign of the great fleet that we had come out to

meet, and almost it seemed that in spite of our certain information as to its course we had missed it, since already we were some hundreds of miles out to sea. Then suddenly, as I gazed ahead, I caught my breath, and the next moment had turned swiftly to the distance-phone.

"Squadron 1 reporting," I said rapidly. "The scouts of the European Federation fleet are in sight and are heading toward us!"

For there ahead a great line of dark dots had appeared suddenly in the empty sky, a great fringe of dark dots that were rushing toward us and that were becoming quickly larger! With each moment that they raced toward us they became larger, until they had come plain to our eyes as long torpedo-shaped cruisers like our own. They differed from our own only in that their bridge-rooms, instead of being raised like our own, were sunk flush with their upper-surfaces, only their transparent forward-windows showing. They were the scouts of the European fleet, and at the same time I saw them they must have seen us, for they changed their course slightly. So racing straight toward us were five hundred cruisers opposing the two hundred of our far-flung line. On and on they came, and I saw momentarily far behind them a great cloud of other cruisers, the mighty main body of the European fleet. I shouted the information into the distance-phone. Then the next moment the speeding line of cruisers before us had rushed straight into our own onrushing line!

The next moment all the air about us seemed filled with whirling, striking cruisers, as the two scouting lines met and crashed. In that first moment a score of our cruisers crumpled and collapsed in headlong collisions with European cruisers. And then as Macklin threw the wheel up at my hoarse cry, our own ship heeled over with sickening speed to avoid two European cruisers hurtling straight toward us. Then as we rushed by them there came the swift sharp detonations of their great heat-guns and a storm of shining cylindrical heat-shells rushed from them toward us. At that moment Macklin swung our cruiser back upward and over the two rushing European ships, and as there came a word from Hilliard to the crew, our own keel heat-guns rained down a score of heat-shells upon the two ships. One of those ships the heat-shells missed, but the other was struck squarely by three of them.

Instantly there was a blinding flare of white light as the striking heat-shells burst, releasing upon the luckless European ship all the terrific heat contained within them, the vast vibrations of radiant heat. For this was the most deadly weapon of modern air-warfare, these shining shells in which, by special processes, the vibrations of intense radiant heat could be concentrated. And as those shells struck and burst upon the luckless ship below we saw the ship hang motionless for a moment in the midst of that blinding flare, its metal sides glowing and fusing. Then we saw it plunge downward like a great meteor toward the gray Atlantic!

But now our own cruisers were whirling up and backward, back toward the struggling ships that hung now in a mighty, struggling line. Like swooping hawks our own craft flashed, diving down upon that battling line with bow and keel guns raining heat-shells upon the European ships below, racing down at a giddy angle into that wild melee of struggling ships and heat-shells that the combat there had become. So wild and fierce had been the combat in the few moments since we had met the European scouts that already scores of ships had plunged down in white-hot destruction toward the ocean. But we had, I saw, well accounted for ourselves in those moments, since almost twice as many of the European cruisers had fallen as our own, and they seemed staggered. Then as our ships leapt like angry birds of prey after them, there came a quick order from the distance-phone that abruptly halted us.

"Main body of European forces approaching! All front and side scout-squadrons rejoin our fleet!"

Trapped!

Instantly Macklin whirled our cruiser again up and back, and as the rest of our scout-squadrons turned and leaped back through the air after us, we saw that the battered European scout-lines were receding also, racing back toward their own main fleet. That mighty fleet was in full sight to the eastward now, its five thousand great cruisers advancing majestically toward us in the familiar battle-formation of the European Federation—a great ring or hollow circle of ships. On they came, the scouts taking their

place within that circle with the rest. Then we, too, had fallen back into place at the head of our own two great columns, the silver-striped flagship of the First Air Chief before us, and slowly now, with ten miles more of clear air between them, the two giant armadas were advancing toward each other.

Standing there with Macklin, heart pounding, I gazed watchfully ahead as our fleet and the European one swept nearer toward each other. We came each withholding our fire for the moment, since the heat-guns have but a short effective range. Although outnumbered two to one, we were moving steadily toward the oncoming giant circle of the enemy. Then suddenly the ships of the great European fleet, still holding its circular formation, had leapt steeply upward with sudden tremendous speed, to slant above us!

As they did so, a quick order rang from the distance-phone and the two great columns of our fleet had leapt upward also, up to the level of the other until a split-second more would have seen us crashing headlong into that oncoming circular fleet. I saw the air before me filled with gleaming ships rushing lightning-like towards us, heard another order ring out, and then Macklin had swung our cruiser to the right and our whole great fleet had divided, one column flashing like light to the right of the oncoming European fleet and the other column to the left of it. Before they could change formation or slant down to escape that swift maneuver of ours, we were flashing past them on both sides, and then to right and left of them our heat-guns were thundering and loosing a storm of swift heat-shells.

As those shells struck, as our passing column loosed a hail of them upon the European cruisers, the air about us seemed filled completely with blinding bursts of light and heat. Scores, hundreds, of the enemy ships were withered by that deadly fire from right and left, glowing and melting and plunging downward like chariots of white fire.

Surprised as they were by our swift maneuver comparatively few fired upon us as we raced past them, but even those few shells found their marks among the cruisers of our rushing column. Cruisers of my own squadron were struck and hanging there glowing and fusing from the terrific heat released upon them, unable to avoid the fast-speeding ships behind them which raced headlong into the white-hot wrecks. Then our columns were

past them and as behind us their ships fell thick in white-hot melting ruin, I turned toward Macklin, exultant.

"We're beating them!" I cried. "Another blow like that one and——"

A cry from the second officer cut me abruptly short, and quickly I gazed back to where he was pointing, toward the mighty ring of the European fleet. Our two columns had converged inward toward each other after that deadly blow, when the great ring-shaped formation of cruisers behind us had halted abruptly its own forward flight, and had shot back a great double file of its cruisers between our own two racing columns! And then, before we could see and forestall its menace, before we had time to obey the swift command that the First Air Chief shouted from the distance-phone, that double tongue of ships had split, each line moving sidewise with terrific force and speed toward our own two lines, pressing them outward from each other, separating them, rolling them sidewise and backward in two great enveloping motions.

In that moment I felt our cruiser reel madly as a European cruiser shot against it, saw Macklin clinging madly to the wheel as I was thrown down and backward, while about us in that mad moment the heat-shells were speeding forth from ship to ship to burst in flaring destruction about us. Then as Macklin swung our cruiser up to a level keel, our heat-guns beneath detonating now as our gunners worked them like mad beings, we were fighting the remorseless lines of the enemy that swept us back and I was aware that our fleet's two columns had been swept hopelessly apart, that our forces had been fatally divided and that each division of them was now completely encircled by the outnumbering masses of cruisers of the European fleet!

Cruisers on all sides of us now seemed to fill the air, enemy cruisers that tossed about us in a great sea of ships and that made our own ships the target now of their unceasing volleys. Our column, rolled together by that irresistible maneuver, had massed into a solid group, the silver-striped flagship of the First Air Chief just beside our own. The air around us was livid with flares of blinding light as the heat-shells broke and burst in unceasing destruction, as the thunder of our detonating guns seemed to drown all other sounds in the universe.

Not for long could we thus remain the target of these masses of cruisers that swarmed about and above and beneath us. Our other column had been swept back and that was surrounded by enemy cruisers and fighting desperately even as we were. Unless we could join them, and reunite our shattered fleet, we must inevitably be destroyed. At that moment the voice of the First Air Chief rang from the distance-phone before me in a high command.

"Triangle formation!" he shouted. "North at full speed!"

Instantly the ships behind and about us, reforming swiftly and smoothly even under the rain of shells shifted into a great wedge-shaped formation, a great triangle of solid ships whose apex was the First Air Chief's cruiser, and which pointed north, toward the other isolated and struggling half of our fleet. Then the next moment our great triangle had leaped forward straight toward the north at full speed, into the swarming masses of European ships that surrounded us. Our own cruiser hung just behind the First Air Chief's, just behind the triangle's apex. Then with a terrific crash we had smashed into the solid wall of ships before us.

Our cruiser rocked and reeled beneath me as its sharp stem rammed at full speed into a European cruiser that had swung broadside in an attempt to escape us. Its side crumpled beneath that awful blow and I saw it reel back and downward, I felt other rending crashes that shook our ship wildly as our triangle crashed through the European fleet. Then suddenly we were through it, had smashed our way by sheer force through its sea of ships and had reached the second half of our fleet, joining with it once more. Scores, hundreds even, of our own cruisers and of the enemy's were tumbling and twisting downward toward the sea, battered wrecks of metal that had been all but annihilated in our mad crash through the enemy armada!

Now swiftly our re-united fleet, still almost two thousand strong, were massing together in a single long rectangle, our flagship speeding to its head, and as we moved toward the scattered swarms of European ships about us, that numbered almost four thousand still, they had formed into a similar formation. Then as our own long rectangle or column rushed toward them they were racing sidewise at the same speed as ourselves, so that side by side now our two great fleets sped through the air, our heat-guns detonating again as we held still to the awful struggle. Our cruiser seemed

to bear a charmed life, since as we drove headlong through that hail of shining death, behind the First Air Chief's cruiser, we were sometimes missed by inches only. And now as Macklin, his eyes steady but burning, held our ship onward with those about us in this mad running fight of the two great fleets, I was aware that in that fight they were both slanting steadily *downward*, down toward the gray Atlantic far beneath!

Fleet hanging to fleet, the air between them thick with shining heat-shells, down we rushed until we were within yards and then feet of the ocean's tossing surface! But, still firing at each other steadily, they were swooping downward still until we were plunging straight down into the ocean's depths. For these great air-cruisers could move beneath water as well as through the air. Each opening in them sealed tight during flight, their air-supplies always automatically furnished by great tanks of liquid-air, their great tube-propellers sucking water through them at immense speed even as they did air, and hurling the cruiser on at a speed which while far less than that in the air was still great—with these features our cruisers were now down into the great waters of the Atlantic.

"Hold steady!" I cried to Macklin as we swooped downward, and the waters rushed up toward us. "Keep in line with the First Air Chief's ship!"

I saw his hands clench upon the wheel, and then the waters were just beneath us, were rushing nearer and nearer, while even then our ships and those about us were loosing their heat-shells upon the European fleet whose great column was plunging downward like our own. Down—down—and then with a shock our cruiser had plunged into the great waters, had rushed beneath the waves, and instantly the light of sunset all about us had vanished, had given way to the green translucence of the waters. Through that green obscurity there shot yellow shafts of revealing light, the underwater searchlights in the walls of our cruiser which I had snapped on. From all the ships before and behind us came other brilliant shafts. Our great fleet still grappled with the European fleet rushing down to our right, our heat-guns loosing their deadly shells still through the green waters toward each other's fleet! The great battle over the Atlantic was to be carried on in the great ocean's very depths!

CHAPTER III

Under the Sea

Green depths that swirled about us, shafts of yellow light that swung and stabbed through them, rushing cruisers and detonating guns and drone of motors and wild shouts—all these merged and mingled in one great phantasmagoria of strange impressions in those first moments. I had shot under the ocean's surface in my cruiser many a time before, but never in battle. And now, with our two great fleets plunging down into those peaceful depths, all about me seemed for a moment a strange dream. Then I saw before us, the cruisers of the First Air Chief and those about him, dark long bulks that gleamed there in the depths beneath us as the yellow shafts of light struck and crossed them.

Peering downward, figure tensed over the wheel, Macklin was holding the cruiser behind those rushing ones ahead, and now, looking away to the right, I could make out the dark, long bulks of the European cruisers also. And across the gap from fleet to fleet were hurtling storms of the heat-shells still, shot forth by our great heat-guns whose valve-breeches made them capable of underwater operation. And as they burst there broke from them the same great flare of light and heat as in the air above, little affected for the moment by the waters about them, destroying in that moment the ships they struck and making the waters about those fusing ships boil terribly with their terrific released heat.

But straight downward through those boiling waters swirled and swept the following cruisers of the two great fleets. As our guns thundered there in the great deep, as heat-shells raced and broke and flared about us, I saw schools of fish and strange sea-creatures and denizens, for a moment in the glow of the yellow searchlights or the flares of bursting heat-shells. The fish were all striving desperately to escape from this hell of battle and death that we men had carried down with us. And still downward—our two great columns

were racing, hanging to each other with fierce, resistless tenacity, raking each other still with the great heat-guns as we shot lower into the mighty depths!

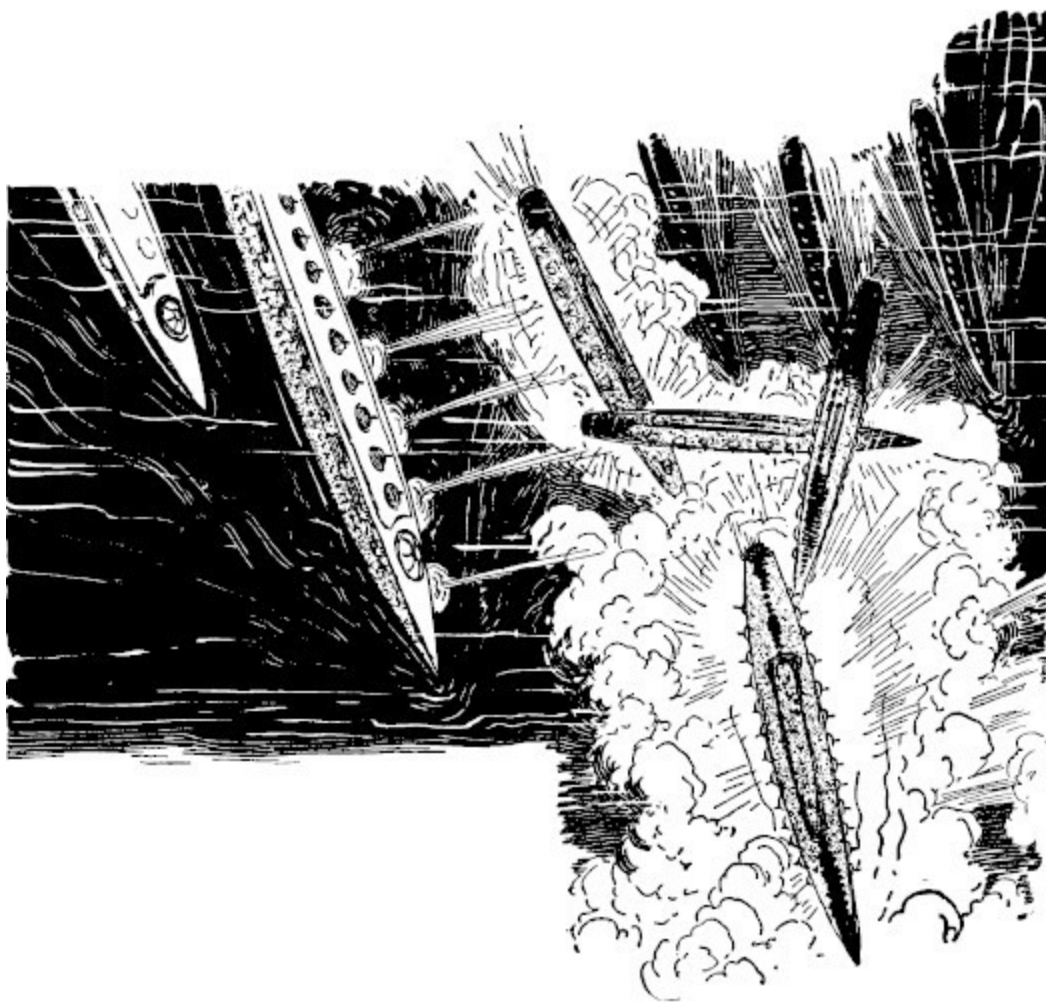
Finally Hilliard dashed up into the bridge-room from below.

"This can't keep on much longer!" he cried. "The cruiser's walls can't stand this heat and speed!"

"It'll have to keep on as long as the First Air Chief keeps on!" I shouted to him, over the drone of motors and thunder of guns. "If the battle is to end for both fleets here—let it!"

But I saw even in that moment that Hilliard was right, and that the walls about us, the transparent metal of the windows, had become searing to the touch. Not only had we raced through areas of water boiling at terrific temperatures from heat-shells that had burst in ships there, but our own immense speed was producing by its friction with the waters a heat that was almost softening the cruiser's walls. Yet I saw that still the First Air Chief's cruiser was rushing deeper and deeper before us, and that still the great column of our own fleet and that of the European fleet were following locked in that colossal death-grip, their heat-guns thundering still toward each other.

I could see too that the cruisers of the European fleet were suffering far more than our own in this awful undersea battle, since there in the green depths, only able to half see each other and to aim their heat-guns by the uncertain light of their searchlights, their greater numbers were of but small advantage to them. And our gunners, following the former orders of the First Air Chief, were concentrating their fire upon the European column's head, so that when ships were struck there by heat-shells, changed to motionless white-hot wrecks in the waters, those behind were unable in the green depths to see them in time to swerve aside, and so crashed into the fusing wrecks and were themselves destroyed. It was a maneuver that the First Air Chief had long before explained to us for use in undersea warfare, and now it was proving of the highest effectiveness and score after score of the European ships were flaring and crashing in their opposing column.



Our gunners, following the orders of the First Air Chief, were concentrating their fire on the European column's head, there in the ocean's green depths.

For only a moment more, though, did the two great columns continue thus, for then the European fleet, feeling the great losses which it was experiencing in this terrific underwater combat, responded suddenly to some order, curving sharply upward again. Instantly the First Air Chief snapped an order from the distance-phone, and instantly our own great column of ships had turned upward too, had curved upward through the waters after the racing European fleet like wheeling sharks after prey, their guns and ours still beating a tattoo of thundering death there in the great

depths. Now as we rushed upward again at undiminished speed the waters were becoming green and translucent once more. Then as we flashed up through those green depths, heat-guns sounding still from fleet to fleet, the cruisers ahead and above us, and then our own, burst suddenly up from the waters into the sunlit air once more!

Into the Clouds

Surely some battle out of a nightmare was this, in which our two great masses of cruisers hung still with deadly purpose upon each other. Macklin and Hilliard and I aware of ourselves now only as infinitesimal and unreasoning parts of this mighty fleet about us, moved upward, miles again above the waves. The two rushing fleets slowed, halted, as though by mutual purposes. Slowed and halted there in two great masses of cruisers in mid-air, our own to the west and the European one to the east, and then, with every heat-gun detonating and with the air between them seemingly filled with shining, hurtling shells, they were hanging motionless in a mighty death-grip!

The great struggle for its sheer intensity was appalling, as the two giant fleets hung there unmoving, high in the air, each unheeding its own danger, intent only upon annihilating the other. I was aware, as though I were a spectator, that I was shouting hoarse commands into the order-phone, that in obedience to those commands our gun-crews beneath were working the great heat-guns like madmen, loosing an unceasing hail of shining shells toward the fleet opposite, shouting as they did so even as Macklin and Hilliard were shouting wildly beside me. I was aware of heat-shells that seemed exploding all around us, of brilliant and unceasing flares of blinding heat and light that burst in dozens each second amid either fleet, their cruisers whirling downward now in score of hundreds.

I know now that that motionless battle there in mid-air could not have exceeded a few minutes, yet then it seemed an eternity. I was aware dimly that our ships were falling faster than the Europeans, that their greater numbers were telling upon us once more here in the open air, and that but few more than a thousand ships were left to us, no more than half of our

original number. Yet more than twice that number of European cruisers remained, smothering us with shining shells! Then suddenly the silver-striped ship of the First Air Chief, that had swayed beside our own, turned westward, and at the same moment Yarnall's voice came sharply from the distance-phone.

"Retreat-formation!" he was shouting. "All ships retreat westward at full speed!"

"Retreat!" My cry was one of incredulity, of mad anger. Retreat—we were beaten, then, our great battle lost—I was aware of Macklin hovering in irresolutely over the wheel, of Hilliard almost sobbing in his rage.

Then despite our fury the sense of discipline was reasserting itself, and with the First Air Chief's ship at its head, our great mass of ships was turning, was forming swiftly into a great T, the longer column or stem of it pointing westward, moving westward at swiftly mounting speed with the flagship at its head, while the shorter column or head of the T lay across its rear at right angles. This protected us somewhat from the European fleet that now was leaping swiftly after us, triumphant, exultant at our flight. Our stern guns still firing toward them as they leapt upon our track, we raced westward, on until at full speed. And now, even as the thunder of guns still came to our ears from behind, a dull, dead silence reigned over our own ship, and those about us, Macklin and Hilliard as silent beside me as myself, a silence of the apathy of utter dismay and despair. For never, surely, had any American fleet ever thus fled homeward, before, pursued by a conquering enemy.

On to the westward though we raced still, our rear-guard line of cruisers now the targets of numberless heat-guns. Still cruisers among them were being destroyed by the heat-shells, and still, too, they were striking savagely back to find their marks here and there among the mass of our pursuers. On and on we rushed, the European fleet closing gradually toward us, and now we were but a score or so of miles from the coast, I knew, and should be sighting the great double line of our air-forts that were hanging far out to sea. It was the one chance of escape for our outnumbered fleet, I knew, to gain the shelter of those great forts. And now it was clear that it was with this object that the First Air Chief was leading our fleet in full retreat westward. But as we gazed ahead, we saw that though we should have been within sight of them the great air-forts were nowhere to be seen!

Save for a great, long bank of floating white clouds ahead the sky was completely empty, and of the air-forts there was no trace!

"The air-forts gone!" I cried. "Our last chance gone!"

"But our fleet's going on!" exclaimed Macklin. "The First Air Chief's leading us into those clouds!"

The Ambush

Gazing ahead, incredulous, I saw in a moment that it was so, that the First Air Chief's cruiser was flying straight on toward the great long bank of clouds ahead, leading our whole fleet into their fleecy white masses. Even as I stared unbelievably, I saw his silver-striped ship rush into those clouds and vanish from view, and after it were rushing our own ship and all those about us, all the long mass of our fleet! Unable to credit my eyes, almost, I stared, for it was a suicidal maneuver, to attempt to elude our pursuers in those fleecy masses. They needed only to surround the cloud-bank and then wait and destroy us one by one as we emerged again. Yet even as I gazed forward our ships were speeding into the white masses of vapor, after our flagship, our rear cruisers still returning the fire of our pursuers. Then as our own cruiser flashed into them, all things vanished from about us save the thick masses of cloud-vapor that hemmed us in, that seemed to press against our windows, curtaining all things else from sight!

I stared forth tensely with Macklin and Hilliard in a vain attempt to see through those masses, heard the thunder of guns still going off blindly somewhere in the great cloud-mass behind us, knew that in the wild heat of pursuit the European fleet had rushed after us into that great cloud-bank. Then came a swift order of "All ships halt!" from the distance-phone, and as we came swiftly to a halt there in the blinding, fleecy masses, motors droning still, we heard the crash of ship on ship behind us in the cloud-bank as the foremost cruisers of the European fleet drove blindly into our own, then halted fearfully themselves, milling confusedly about in fear of further collisions and with neither fleet firing now in the absolute blindness that held each ship. Thus the two mighty fleets hung there for the moment blind

and helpless in the huge cloud-bank, and in that moment there came again the First Air Chief's voice from before us in a swift, shouted command.

"All American Federation ships—*drop!*"

Before the order had ceased to echo Macklin's hand had flashed to the power-stud, and as the great drone of our motors suddenly lessened our cruiser dropped downward like a falling stone, plunged downward until in a moment more it had ripped through the great fleecy mass of the cloud-bank and into the open clear air beneath it, leaving the great European fleet for the moment still in it. And in that moment, even as our cruisers halted their plunging downward fall, there came a great hissing sound from above as of the hissing of terrific jets of air, and at the same instant we saw the mighty cloud-bank above breaking up, disintegrating, its great fleecy masses whirling suddenly away in all directions, driven away in a moment as though by mighty winds, breaking away in formless flying vapors! Breaking away to leave clear air where they had been, to leave the European fleet hanging there, appearing to our sight suddenly as a confusedly milling mass of numberless ships above us! And coming to ——? on either side of that confused mass of ships was *the great double line of our giant air-forts!*

"The air-forts!" My cry was echoed in that moment by Macklin and Hilliard beside me, by all in our cruiser, in our fleet.

The air-forts! On either side of the disorganized European fleet they hung, in their mighty double line, and as that fleet saw them now for the first time with the sudden disappearance of the cloud-bank that had hidden them, it seemed to hang motionless still as though stunned with astonishment. Then the great heat-guns of the air-forts had swung toward them, were thundering in swift chorus, were loosing storm upon storm of heat-shells upon the confused, astounded ships that swung between them! Were pouring forth in that awful moment all the concentrated fire of their mighty batteries upon the European ships caught between them.

The air-forts! And it was between them, between their two mighty lines, that the First Air Chief had purposely led the European fleet, I saw now. For *this*, then, was the new device of the air-forts of which he had spoken to us before our start, this device which enabled them to surround themselves with a great cloud-bank that kept them hidden from all and unsuspected by

any enemy. Some device for projecting forth great masses of water-vapor it must be, that had enabled them to form that great artificial cloud-bank about themselves. And when the First Air Chief, staking all upon the device, had led the pursuing European fleet into that great cloud-bank, into that giant ambush of the air-forts, then with our own fleet dropping down out of it they had needed only to disperse the artificial cloud-mass about them by means of great air-jets of terrific power, to disperse the cloud-mass and to turn all the fury of their great guns upon the European fleet that hung still dazed there in the withering fire of those suddenly-unmasked batteries!

For now above us the European ships, whirling aimlessly about in that terrific fire that raked them from either side, were falling faster still! Their own shells burst and flared along the sides of the great air-forts, but were too few in number to cripple or destroy any of those gigantic, heavily-armored edifices. And at that moment, even as the European ships strove to mass together to escape from that great death trap of the air, the First Air Chief's ship was slanting up toward them, and now we needed no orders to follow as we raced up after him. Up until our great fleet rushing upward in a single mass was pouring up before us a third terrific fire of heat-shells which, added to that of the air-forts on either side, sent blinding death-flares dancing and leaping over all the mass of ships above us.



And now above us the European ships, whirling about aimlessly in the terrific fire that raked them from either side, were falling faster still. And even as they massed together to escape that great death trap, we were slanting up after them.

"They're turning!" cried Hilliard. "They're fleeing!"

Homeward

Fleeing! Even as our fleet shot up toward them the European ships, reduced now to hardly more than two thousand in number, and unable to bear the terrific fire concentrated upon them from three directions, were soaring frantically upward above the air-forts, up and away to the eastward, massing together in a close-bunched, irregular formation. And our fleet had shot up after them, sending a rain of shining messengers of death among them as we shot after them, pursuing them with bow-guns firing just as minutes before they had pursued us. Then, broken and disorganized and incapable of further resistance for the time being, the great European fleet was drawing away from us as an order from the First Air Chief halted our wild pursuit. Outnumbered still as we were by two to one we could not carry the pursuit too far from our supporting air-forts.

As we halted, we saw the European ships racing on in a struggling mass, dwindling and vanishing from us quickly against the gathering dusk eastward. Then our own battered cruisers were turning, heading back westward, back toward the brilliant, waning sunset, and with our flagship at our head until we paused above the air-forts. There, with the wild exultation of victory we three in the bridge-room, Macklin, Hilliard and myself, and our crew and all the cruiser crews about us, expressed ourselves in great roaring shouts. And then, once more, there came from the distance-phone before us the voice of the First Air Chief.

"Cruiser-captains and men of this fleet," he said, "we have beaten back the first attack of the European Federation fleet. And I have received but now a distance-phone message from the Second Air Chief, commanding the western fleet out of San Francisco. He reports that his own fleet, meeting the oncoming Asiatic Federation fleet, was able after a battle as terrific as our own to drive it back also, by using the same cloud-ambush device in the air-forts as we used here. Thus on this day to west and east we have accomplished the impossible."

He paused, and at his words, his news, a wilder cheer went up from all our ships and air-forts, hanging motionless there against the crimson of the dying sunset. But now, his voice solemn, the First Air Chief went on.

"We have won today, in east and west, but what we have won is but a respite. The mighty European and Asiatic Federations have gathered all their forces to annihilate our American Federation. Their great fleets have

been cut in half by these two battles, but so have ours. And they not only outnumber us still by far, but they can build new cruisers faster than we. Undoubtedly within weeks, days perhaps, there will come another mighty onslaught from them, from west and east, an onslaught for which they have been preparing and are preparing some colossal and terrible plan or weapon of which we know nothing. It is some unknown device that it is rumored will enable them to move gigantic forces upon us. We must stand against them, nor can we hope to surprise them with the cloud-ambushes used by us today. Yet whatever forces they bring against us, whatever giant new weapons or terrific attacks they loose upon us, whatever is the great end of this Last Air War that today has started, you of the American Federation fleet can be proud always of the way this first battle was fought and won!"

There was silence a moment and then another shattering cheer. And then, the First Air Chief's cruiser leading, our fleet was moving smoothly westward toward the sunset, and toward New York. As we moved on our watchful patrols were already out from the fleet's main body to north and south, while behind us the great air-forts, slowly and ponderously, were following us, spreading into a long single line which with the ceaseless patrols was to guard us from any surprise attacks or raids. Already, by now, the dusk was gathering behind and about us, the sunset's light waning in the west. And by the time that our fleet came again in sight of New York the great air-city's outline was visible only as a mass of brilliant lights floating high in the gathering darkness. The mighty city, as we learned, had begun to move eastward to meet us upon hearing of the results of the day's battles, and now glimmered before us like a great mass of brilliant gathered stars, the giant beams of its searchlights sweeping the night.

Onward and down toward the mighty city shot our fleet, and as Macklin and Hilliard gazed down with me we saw the cruisers that landed upon the white-lit plazas across the immense floating city surrounded at once by joyful crowds, their weary crews carried high on shoulders. The whole great city, indeed, was rejoicing, though that rejoicing was not extravagant, being tempered by the knowledge that it was but the first attacks of the European and Asiatic Federations and that other and greater attacks might be expected to follow soon. So although the great city blazed with lights as our fleet slanted down toward it, its great towers and pinnacles and pyramids seeming like magic palaces of radiance floating there in the night of the

upper air, yet its great watchful searchlights stabbed and circled still, and there came and went still high above it and to north and east and south the humming patrols, on guard now and challenging every craft that approached the city.

Then our cruiser was landing, and Macklin and Hilliard and I were emerging from it with our crew, mindless of the shouting crowds that surrounded every landing plaza, stumbling in our utter weariness through those crowds to our barracks, to fall into a stupor-like sleep of utter exhaustion....

The Respite Ended

It was the middle of the afternoon when we awoke, more than a score of hours later. Our quarters lay in one of the uppermost levels of the great barracks-tower, and as I rose and after dressing joined Macklin and Hilliard at the window, we could see far out over the air-city's great expanse. Above us blazed the afternoon sun shining on numberless patterned windows of all the gigantic metal towers about us. Far overhead there still hummed and flashed the ceaseless patrols, still watchfully hovering above and around New York. Beneath, on the city's landing plazas, there rested still the hundreds of cruisers of our returned fleet, and now we saw that upon the great central plaza where our own ship lay there were gathered now some two hundred and fifty of our twelve hundred and fifty ships, and that about these central ships were swarming a great horde of mechanics and attendants; caring for and inspecting their great motors, filling the liquid-air tanks that supplied constant breathable air, refilling their magazines with shining masses of heat-shells.

I turned puzzled toward the other two. "Strange that they should be giving such swift attention to those two hundred and fifty cruisers," I said.

Macklin nodded, frowning. "And our cruiser among them," he commented. "One would almost think that—" He stopped short as our door snapped open and an attendant stepped inside, saluting.

"Captain Martin Brant to report at once to the First Air Chief's headquarters in the tower," he said, "and all cruiser officers and crews of Squadrons 1 to 4 to rejoin their ships at once!"

Again he saluted and disappeared, leaving us staring blankly at each other. Then we were struggling into the tight black jackets of our uniforms, were striding out in a moment and down to the great air-city's "ground" level in one of the building's electrostatic-motored cage-lifts. Through the crowded streets we strode, seeing now that in all those streets other black-uniformed men of the squadrons named were pressing toward their cruisers in the central plaza. Then we three had reached that central plaza, from whose center rose the mighty electric power-tower, and around which the two hundred and fifty cruisers rested, all of our first four squadrons that had survived the battle.

Already, I saw, the crews of those cruisers were taking their places within them, and as Macklin and Hilliard took up their positions in our own I strode on across the plaza toward the huge tower's base, in which were the headquarters of the First Air Chief. Passing challenging guards at its door, I passed through a few narrow white-lit ante-rooms, and then had stepped into the great circular room that was his innermost office. The curving walls of that room were covered with panel after panel of instruments and switches, which controlled the vast electrical currents that rushed down from the electric-tower's tip and transformers to those motors in the city's base. Near the room's center was the battery of six great switches which controlled the city's direction of motion, moving it in any direction at will at slow and ponderous speed, the speed-control's gleaming knob beside them. And beyond the controls of the great air-city, there stood a great table-map of the world, upon which a myriad of red circles automatically showed the position of the world's air-cities.

Behind this table-map, as behind a desk, the First Air Chief was sitting as I entered, while around the panelled walls there moved a half-dozen black-jacketed attendants constantly watching and controlling the flow of current from the power-tower's tip to its motors. The First Air Chief, as I entered, motioned me silently to a metal seat before himself, at the great table-map's edge, and then for a moment contemplated me in silence, as though considering his words before speaking. Regarding me intently, he began.

"For a second time, Captain Brant," he was saying, "I have summoned you here to me, but this time alone, and with the two hundred and fifty remaining cruisers of our first four squadrons summoned also outside. You are wondering, no doubt, why I have done so.

"The victory we have gained is, as I said, but a respite. We know that the two great Federations, though beaten back with great losses will soon be launching another and a far greater attack upon us, one against which I think we cannot stand. From the European Federation to the east and from the Asiatic Federation to the west that mighty second attack will be loosed upon us, with some terrible new weapon or plan whose nature we cannot guess. For though hundreds of agents have been sent by us to all the European and Asiatic air-cities, months before the outbreak of this war even, they have been either captured and made away with, or have been able to report only that immense preparations of some sort are going on in those cities, in Berlin and Peking especially. And the rumors which have reached us through them indicate that whatever great new colossal weapon or thing they are devising at Berlin and Peking, it is one which, they boast, will enable them to sweep all our cities from the air in a single mighty attack.

"You see, then, that to wait for them to develop their great weapon or plan, to await this terrible attack without action, is but to pave the way for our own doom. We must strike out to halt them, to cripple or destroy their great secret plans, must strike at the European and Asiatic Federations both before they expect us. And that is why I have called you here to me. For it is my intention to launch a great raiding attack of our own at both Berlin and Peking. If we can strike a smashing blow at those two air-capitals, can damage or destroy the great military preparations within their arsenals, which must hold their great secret also, we shall have crippled, for the time being, their plans and shall have gained time for us to learn and counteract those plans. Even now our two hundred and fifty ships are ready and wait to start for Berlin, while from San Francisco a similar number will raid westward to Peking. And it is my order that you, Captain Brant, shall command this great raid eastward, for your conduct in the great battle of yesterday proves you worthy of the command. So soon after that battle, our enemies will never dream of our lesser forces attacking them, so now is your great chance to strike back at them, to flash across the Atlantic in a

great surprise raid and strike down out of the night with all your power at the great air-capital of Berlin!"

CHAPTER IV

A Desperate Plan

For a moment, I think, I stood in stupefied silence as the meaning of the First Air Chief's breath-taking plan sank into my brain. Then I had snapped to sudden attention, saluting, my eyes shining. Yarnall was smiling, too.

"The plan is bold enough," he said, "but it means a chance to strike a terrific blow at our enemies, to cripple and perhaps destroy their great preparations that mean doom for us. The two hundred and fifty cruisers gathered here in the central plaza have been completely replenished with supplies and inspected while you slept, their magazines filled with heat-shells, their bomb-slots with mighty heat-bombs. You can thus start at once, heading straight across the Atlantic toward the air-city of Berlin. And if you can reach it with your cruisers, under the cover of darkness and the unexpectedness of your coming, win through their great patrols and chains of air-forts, and reach the great air-capital, you will be able to strike a blow that may yet save us. I know, and you know, Captain Brant, what perils lie between your cruisers and their goal, but I need not speak of those perils and need not tell you what hopes depend upon your raid. I need only give you now a single order—to start at once!"

Five minutes later our two hundred and fifty cruisers, humming like a great swarm of bees, were rising up into the brilliance of the sky. My own cruiser leading, the familiar figures of Macklin and Hilliard again in the bridge-room beside me, I wondered momentarily if ever I was to return to New York. The mighty city floating there beneath us, its crowds now watching in wondering silence as we rose from it, its masses of buildings suspended there between earth and sky like a strange new galaxy of stars—it was home to me, and it was somberly enough that I watched it dropping now away from our ships.

Upward we rose, hovered, then shot toward the west, driving smoothly until the great mass that was New York had dropped out of sight behind us. Then as I spoke an order into the distance-phone our ships turned, circling widely to the south, and then moved eastward, out of sight of New York. It was a necessary maneuver, I knew, to make it appear that our cruisers had gone westward. Necessary because in New York's millions there were certain to be European spies who would have endeavored to warn their capital had they suspected that we were in reality racing eastward.

And now as we shot out over the Atlantic again, I gave another order and our two hundred and fifty cruisers massed quickly into a compact triangle with my own ship at its apex. It was the best formation for a raiding party, and holding to it our little fleet shot upward now and onward, onward until we were racing above the great line of our air-forts hanging miles out over the Atlantic in a great watchful chain. We had answered their challenge and were rushing on above and beyond them.

Within minutes they had vanished behind us, and our cruisers were rocketing forward at swiftly mounting speed, racing onward and upward until at more than a thousand miles an hour we were rushing eight miles above the ocean's surface.

As we were rushing toward the east, as fast as the sun was rushing away from us, the night came upon us swiftly. There came dusk and then the stars. We were at an altitude at which we would be sighted by almost no other craft, I knew, an altitude rarely used by any ships. Though the modern closed-construction and air and heat arrangements of air-craft made flying at that height practicable enough, it was necessary by reason of the greater tenuity of the air to use more of the motors' power to attain the same speed. As we hummed on at that great height, all sight of the ocean beneath was hidden from us by the great vapor-layer that lay over it beneath us and only the pale stars above and the triangle of gleaming cruisers behind were visible to us. Yet as we shot on, it was not these, our immediate surroundings, that held my thoughts, but the object of our flight. Gazing beside into the night, with Macklin silent at the wheel beside me and with all our long ships rushing close behind, I could not but be aware in those moments of the desperateness of this raiding attack upon which we were engaged.

To flash across the sea with but little more than two hundred cruisers, to attempt a raid upon the European Federation's mighty capital even while a similar raid was made from westward upon the Asiatic Federation's capital, seemed indeed so desperate as to approach insanity. Berlin was guarded by a great chain of air-forts and patrols hanging over the eastern Atlantic; which held within itself, without doubt, all the great European battle-fleet of thousands of cruisers; which bore upon itself countless mighty batteries of giant heat-guns. Could we, in the face of these, reach Berlin, and send our heat-bombs crashing down upon its great arsenals?

Above the Enemy

These were the doubts that assailed me as our triangle of cruisers throbbed on and on through the upper night, but resolutely I thrust them away, remembering what our attack, what the crippling of our enemies' great and mysterious preparations, would mean to our American Federation. Then I turned as Macklin pointed silently to the glowing-figured dial of our distance-log, and saw by it that while I had brooded there at the window we had swept far out over the Atlantic at our tremendous speed. Within a short time, I knew, the European coasts would be beneath us, but during all the course of our flight so far we had sighted no other ships whatever, all merchant-traffic over the great ocean having been swept from the air by the first alarms of war, while we were still too far to the west to be meeting the far-flung patrols of the European Federation forces.

Soon, though, these would be coming into sight, I knew, and the result of our daring expedition depended upon our success in passing them unobserved. If we were seen by them, a minute would suffice for the patrols to give the alarm by distance-phone, and then from all the European air-cities ahead, from Stockholm and London and Berlin and Marseilles and a hundred others, numberless patrol-cruisers would be swiftly converging upon us in answer to the alarm. And the European battle-fleet itself, we knew, in Berlin, the air-city we had come to attack, would be swift to answer also, so that never could we hope to win through if we were but for a moment detected.

But still we were rushing westward through the night, my cruiser in the lead, and still as Macklin and I peered intently ahead and below, Hilliard having taken up his station beneath, we could make out nothing but the chill masses of the great vapor-layer far beneath us, and the gleaming, rushing shapes of the cruisers behind us. Then, I peered ahead and down toward the right, with body tense, and in the next moment had snapped out the green guiding light at our cruiser's stern, and had uttered a quick order into the distance-phone before me.

"European Federation patrols ahead and beneath!" I warned quickly. "All cruisers reduce to quarter-speed!"

Instantly in obedience to that order the triangle of rushing ships behind was slowing, each cruiser swiftly reducing speed, the great drone of their motors dying to a steady hum. Moving forward thus, as slowly and silently as possible, I pointed downward, Macklin's eyes following my pointing finger.

"The patrols!" I whispered to him. "There beneath us—moving northward!"

Far beneath us indeed they were, a little circle of moving lights that hung just above the great vapor-layer and that was moving steadily toward the north, from our right to our left. Some twenty or more of those white lights there were, moving smoothly along in the same ring-like formation, and though we could not see the shapes of the cruisers from which those lights gleamed up through the night, we knew that they could be only one of the enemy's westward patrols, flying in the familiar European Federation circular formation. Watching them, Macklin and I unconsciously held our breath, while from our ship and from all the ships behind there came no sound other than the low hum of the motors. Slowly beneath those motors' lessened power our cruisers were moving forward through the upper darkness, while beneath the little ring of lights were still holding toward the north. Our presence far above them was apparently unsuspected by them.

I knew, though, that if they were to turn toward us by any chance the great cone-shaped cruiser-finders which are set in the sides of all war-cruisers and air-forts and air-cities, that we would be detected soon enough, since undoubtedly the patrol-ships beneath carried them also. Those great cone-like instruments, when turned in any direction can detect by means of super-sensitive induction-balances the operation of any electrostatic-motors. Fortune favored us, though, for without dreaming of our existence there

above them the ring of patrol-cruisers, the circle of moving lights, moved smoothly on to the north while we held eastward until they had vanished behind us.

Now as I spoke a swift order we were picking up speed again, our cruisers accelerating once more to their former velocity. I knew that we must be very near the southwestern coast of England. Our course lay high above that coast, taking us along a line that would lie midway between the two mighty air-cities of London and Paris, avoiding both purposely on our great flight toward the mightier air-city of Berlin. Soon, I knew, the great air-fort chain that guarded the whole western coasts of Europe would be drawing within sight, and intently enough we were peering forth in search of it, but though that must be passed still we had won through apparently, the outer patrols, without discovery.

"It's hardly likely that they'd have a second line of outer patrols out," I said to Macklin, as we peered together through the dim night from the bridge-room of the rushing ship. "And once we get past the air-forts we'll have a good chance."

He nodded. "They'll never dream of us making a raid upon them tonight, and if we aren't picked up by the air-forts' cruiser-finders we can reach—"

He broke off, suddenly, and at the same moment as he, I gazed down toward the right. Another ring of moving lights was there in the darkness beneath, northward, too. But this one had paused for a moment and was slanting straight up toward us!

"Another patrol!" Macklin's cry was echoed into the distance-phone.

Another patrol—and it had seen us! And then, even as that patrol's twenty cruisers slanted up toward us, to challenge us, eighty of the cruisers of the lowest of our great triangle of ships had whirled like light down toward them, without command or formation, whirled down upon them massed together like a great striking thunderbolt of gleaming metal! For they knew, without need of command, that in an instant more the patrol-cruisers beneath would see and recognize the purpose of all our racing ships, would instantly with their distance-phones send the alarm spreading like flame over all the European Federation. And so our eighty down-rushing cruisers, massed solidly together, fired no guns and dropped no bombs, but simply

flashed downward in a terrific ramming swoop and in an instant more had crashed their great mass squarely into the ring of the uprising European ships!

There was a rending crash of metal that seemed to split the air beneath us, and then in a great shower of wrecked and twisted cruisers the ships beneath were falling, tumbling down and vanishing into the vapors far beneath on their headlong fall toward the Atlantic! All of the twenty enemy cruisers, and about twenty-five of our own four-score that had crashed down into them, fell thus, annihilated almost by that terrific collision. It had been the one means, though, of instantly destroying their patrols without using our heat-guns whose detonations might give the alarm. And we knew that only that swift, unordered action on the part of our lowest ships had saved us. Then the fifty-five survivors had rushed up again among us, and then our ships that had slowed there for the moment were rushing still on eastward.

The Air-Forts

Onward we shot through the upper night, shaken still by that sudden peril and escape, and then I uttered a warning word into the distance-phone from our cruiser leading. For now, far ahead, we could make out great beams of white light that hung in a great row extended from north to south as far as the eye could reach, and that seemed like white fingers of light whirling and reaching through the air as they ceaselessly swung and circled. A full four miles above the earth, and more than that beneath the level of our own onrushing ships, hung this great line of restless beams, and we knew it, at once, for the great line of air-forts that guarded the western approaches of the European Federation. For the beams we saw were the great beams of the air-forts' mighty searchlights, and those swinging shafts of radiance were of such intense brilliance and magnitude that even at our greatest flying height we could not hope to pass over them undetected.

It seemed, indeed, that to pass them was hopeless, since the air-forts, hanging above the great layer of misty vapor that stretched beneath, could instantly detect with those mighty beams any cruisers passing above them,

at whatever height, and could blast them from the air with their gigantic batteries of heat-guns. To pass beneath the great vapor-layer was as impossible, since the air-fort chain which the European Federation put forth here in war-time was a double one, and its second line hung, farther eastward a little, beneath the vapor area, watching with its own great beams and guns for any ships passing there. There remained but one alternative, to pass through the thick mists of the vapor-layer itself, but that, though concealing us from the guns of air-forts above and beneath, would be in itself suicide, since such vapor-layers between the forts were invariably filled in war-time with floating air-mines, great cube-like metal containers held aloft and motionless by their own electrostatic-motors and tube-propellers and which contained a terrific heat-charge which was instantly released upon any luckless ship that touched them.

But now as our ships slowed at sight of the ominous fingers of light far ahead I spoke quickly into the distance-phone. "Our one chance is to go through the vapor-layer," I said, "and use our cruiser-finders to avoid the air-mines. By going through in a three-ship column we may be able to make it."

At my order therefore our great triangle of cruisers shifted its formation abruptly into one of a long slender line, three ships in width, and then that line with my own cruiser at its head was slanting sharply downward toward the great mists beneath us. A moment more and our cruisers had entered those mists, were moving forward enveloped in them, the great vapor-layer through which we moved hiding all things from about us, hiding our cruisers even from each other. But though we could not see them, we knew that the great air-forts hovered ahead and above us, now, and that the vapor-layer into which we were moving was one sown thick with the deadly air-mines. So, with Macklin at the cruiser's wheel guiding it slowly forward at the head of our column of ships, holding a course eastward through the mists by the compass and creeping forward now at the same low speed as the ships behind, I ordered Hilliard, beneath, to swing out the cone-like cruiser-finders from our ship's sides, and to report instantly any air-mines they detected before us.

Behind us, too, the cruisers that followed were using their own cruiser-finders as they crept through the mists after us, at my order; for though as leading ship we could report to them all air-mines which we encountered

before us, it was necessary for the cruisers behind to feel their way forward independently, since in the concealing mists they could not follow exactly upon our own ship's track. Now, though, listening intently at the order-phone, I waited Hilliard's reports. And in moments more, as our cruiser-finders' coils picked up the hum of the enemy's electrostatic-motors a little ahead and to the right, he reported sharply and I repeated the information swiftly to Macklin, who instantly swung our ship a little to the left. And still Hilliard remained with the cruiser-finders, whose super-sensitive coils caught instantly the electrostatic-motors of the air-mines before and about us.

Onward thus we crept, Hilliard reporting at intervals of every few moments as an air-mine was picked up ahead, while at my swift repetition of his report Macklin would swerve our ship to avoid it. Behind our own craft, we knew, all the scores of our cruisers were creeping forward through the great vapor-layer in the same manner. Now we could plainly hear the great, unceasing drone of the mighty air-forts above, as we crept through the vapor-layer beneath them, and knew that were we to emerge into any chance opening in the thick mists about us we would have but short shrift enough from the giant guns of those forts overhead. Yet still we crept on, praying that none of our cruisers struck the deadly mines, since a single one striking would loose a great flare of heat and light from the bursting air-mine that would betray us all. Even our own ship, as it swerved from an air-mine that Hilliard had hastily reported, almost ran full onto another one in the opposite direction, a great cube of metal, holding within it a hell of condensed heat and death and suspended by its power gained from the concentrated cosmic trap. And though Macklin whirled our cruiser aside in time to graze by it it seemed impossible that all our ships could feel through this field of death without disaster.

Yet still we were creeping onward, through the thick mists, and now the great air-forts' drone came from behind and above us, as we passed on beneath them. On and on, feeling blindly forward through that zone of potential death we went, over the second chain of air-forts whose motors' sound came up to us muffled through the mists, and then that too was dropping behind us. For some moments, though, we continued to feel forward in the vapor-layer, and then I had given the ships behind the order to rise and at once, as carefully as ever, our cruisers were feeling their way

upward until they emerged at last into the open air above the mists, a tight steel hand seeming to uncloset from about my heart as we came up from out that terrible zone of death into the dim starlight of the upper night, the white beams of the upper air-forts now far behind us.

On to Berlin

"Through at last!" I cried to Macklin, as we drove upward. "It seems incredible that all our ships could have won through that mine-field!"

Macklin nodded. "We'd not have made it had the air-forts there been using their own cruiser-finders," he said. "But they never dreamed that any ships would try to get through the mine-sown mists, evidently."

Now I spoke into the distance-phone another order, and our ships were swiftly forming into their triangle formation, were racing forward again at rapidly mounting speed to the east, air-forts and deadly mines and questing outer patrols out of sight. And now, as with Macklin and Hilliard, who had joined us from beneath after his work with the cruiser-finders, I gazed forth, I could see that the great layer of mists beneath us was thinning somewhat as we raced on, knew from that fact that we had raced from above the Atlantic and now were moving far above land, since always these mist-layers were far denser above the sea than above land. That land over which we were now speeding could only be that of southwestern England, I knew, and even now our flashing triangle of cruisers was veering further to the south to avoid the great air-city of London. Then, as we hummed on eastward at the same great height as before, we made out a great mass of lights far to the north, a mass of white lights that hung high above the earth and that glowed toward us like a single soft light through the mists that lay between it and our eastward racing ships, smaller beams stabbing and circling from it.

There were needed not the exclamations of Macklin and Hilliard beside me to inform me of that great light-mass' identity, for an air-city of that size in this region could be but London. The great city, I judged, had moved eastward somewhat from its usual position over the center of southern

England and further away from the great chain of air-forts and mine-fields that guarded it to the west. It was not London, though, that was our flying force's objective on this night, and we raced onward with no backward glances toward it, peering ahead with growing tenseness. Far below us we could glimpse, now and then, occasional formations of merchant-ships flying toward or away from London, and convoyed usually by a half-dozen war-cruisers, but these were far beneath and as we were showing no lights and rushing on at tremendous speed they did not glimpse us.

No patrols were in evidence now about us, the main reliance of the European Federation air-chiefs having apparently been put upon their great outer circle of air-forts and patrols, through which we had managed to break. Nor, was it evident, did they dream that the American Federation, depleted as its fleets were despite their victories in the battles of the day before, would attempt any such daring attack upon an enemy so superior as we were rushing upon now.

As we fled onward, holding our three-sided formation, I wondered momentarily what that other American force was now doing that was heading in the same way toward Peking, and then my wonder passed as another great glow of white light showed itself ahead and to the south. It was Paris, we knew, a great air-city as large as London and outranked in size only by the three colossal air-capitals of the world. But it was not Paris, either, that was our goal, and we veered now to the north somewhat to avoid it, flying on at such a great height and distance from it as to pass far beyond the reach of the great searchlight beams that swung and circled from it as they had done from London. Then it too had dropped behind to the south, and regardless now of the other air-cities that we glimpsed far off in the night, we were rushing eastward high above what had once been France, were speeding forward at the same tremendous height on the last lap of our daring journey.

Now other masses of air-traffic were manifesting themselves far beneath us, as squadrons of moving lights, but neither Macklin nor Hilliard nor I, nor any in our ships, were paying attention to these, all our souls centered on the horizon ahead, on the dim darkness of night that stretched before us. Gazing out into that darkness, my two friends beside me, as tense as I leaned, there at the bridge-room's windows as our droning flight of ships sped on. Nothing dispelled that darkness but the dim starlight from above,

but now, as we gazed forth, we became aware of a faint light coming feebly toward us from far ahead, a faint light that seemed like a great, feebly-glowing cloud in the darkness, and that was intensifying in radiance with each moment that we rushed toward it. The glowing cloud seemed to sink steadily as we sped on, seeming to become lower until from our own ten-mile height we saw at last that it was hanging at a height of four miles from the earth. And swiftly it was growing in size, ahead and beneath us, until as we neared it high above, it changed suddenly to our eyes from a great glowing cloud of light to a colossal circle of uprushing white radiance, a mighty circular city floating there in mid-air, that was as huge as New York itself, and that blazed in the night before us as our own city was wont to blaze.

"Berlin!"

Our three exclamations came together in that moment, exclamations that must have been echoed then from every watcher in our onrushing ships. Berlin! In all its stupendous, radiant splendor it hung before and beneath us, the mighty air-city that was the European Federation's capital and center, equalled in size only by New York and Peking. There between earth and stars it floated, its white-lit towers soaring up from the mighty metal base, all out-topped by the slender central pinnacle that was the great city's electrostatic tower which drew from earth's charge its electric power. Around the city's edge there stabbed and circled the giant white beams of its great searchlights, sweeping to and fro over the still-thronged streets, in which we knew there surged the crowding masses of the great air-city's population. And high above these, moving restlessly to and fro, there came and went the great network of patrols which guarded the great metropolis of the air on all sides.

But our own ships, winging more slowly on at our tremendous height, were never glimpsed by the patrols so far beneath us, never caught at our great height by the great white beams that came and went below, and that only occasionally clove the night above. And as my order brought our ships to a halt, we could make out more details in the white-lit city floating far beneath us. Could make out, as we hung there motionless, the great batteries of pivoted heat-guns set at the central plaza and all around the city's encircling wall, the great square metal buildings of the arsenals, in two groups at the city's east and western edges, the central headquarters and

arsenals of all the European Federation's military forces. On the plazas around those buildings rested long ranks of gleaming cruisers, cruisers that numbered thousands and, we knew, were those with whom we had battled so furiously over and in the Atlantic a day before. And it was down toward these buildings and these cruisers that we gazed now, in that moment before the city's cruiser-finders beneath could detect us and spread the alarm.

"The cruisers and military buildings and arsenals below will be our main objective," I said into the distance-phone as we hung there in that tense moment, above the shining city. "The city's electrostatic tower is so closely defended by heat-gun batteries that we could never get near it, and like all power-towers of air-cities it's of metal alloys that the heat of our shells and bombs wouldn't affect, so we can't hope to destroy it and thus crash the city to earth by cutting off its sustaining flow of power. Our goal must be the cruisers and arsenals, and we'll attack them in two great swoops, the eastern ones first and then the western, and if all goes well can then swiftly escape before the forces below can gather and rise against us!"

A Sudden Attack

Now, poised there miles above the great air-city, which was itself poised high over the earth, our great triangle of ships hung like so many birds of prey for the moment. Beside me Macklin was gazing downward as tensely as myself, Hilliard beneath with our waiting gunners, while under my fingers lay the four rows of white buttons the pressing of each of which would release from our cruiser's bomb-slots a portion of the immense heat-bombs they held. Poising there in that tense moment the whole scene was imprinted unforgettably upon my brain—the gloom of night about us, the vast radiant circle of the colossal air-city beneath, the patrols swarming over it, the throngs that filled its streets, excited no doubt over the beginning of the long-expected war that was to annihilate the American Federation. Then I spoke one sharp order into the distance-phone and instantly with all our motors droning suddenly loud our great triangle of cruisers was diving straight downward upon the radiant air-city beneath!

Down we shot with dizzying speed in that mighty swoop, down with my own cruiser flashing foremost and with all our others close behind it, down through miles of space in a flashing moment, it seemed, until our hurtling wedge of ships had crashed down into and through the swarming patrols above the city, had driven like light down through them toward the eastern mass of military structures and cruisers that was our goal! From all of our ships no single gun sounded nor from the patrol-cruisers through which we dropped, so stunned were they by our great crash downward through them. It was as though for that moment a tense silence had been enforced upon all the world, a silence broken only by the drone of our motors as we plunged. Then I was aware in a swift succession of flashing impressions of the white-lit city's towers and buildings rushing like light up toward us, of the great square military arsenals and buildings with the gleaming ranks of cruisers about them, just beneath us. Then as we plunged to within a half-mile of those buildings and cruisers my own foremost-flashing cruiser curved forward and, as our down-plunging ships levelled out behind it, I pressed swiftly a row of the buttons beneath my fingers. The next moment our cruiser was swaying from side to side as it rushed on, and down from it and from all the massed ships behind and about us were plunging thousands of giant, cylindrical heat-bombs!

Even before those heat-bombs struck, our onrushing ships had curved like lightning upward again, but the next moment were reeling and tossing even in the mad upward rush as from beneath came a titanic merged flare of all the bursting heat-bombs, from which an awful wave of super-heated air rushed up and overtook us, and beneath whose terrific released heat dozens of the huge military buildings beneath had fused and melted. We could glimpse, too, that below a full half thousand or more of the resting cruisers had perished also in that giant flare, and that it was as though a whole great area of the gigantic air-city beneath us had been transformed suddenly by the released heat of our mighty bombs into a huge crater of white-hot, melting metal near the floating city's edge! And from all across the mighty white-lit mass of Berlin, that had reeled itself in mid-air from that terrific blow, there rose a dull, roaring clamor of millions of voices that came up to us even over the drone of our great motors and the rush of winds about us!

Upward at utmost speed we were rushing, and just in time for hardly had our heat-bombs struck when, despite the utter unexpectedness of our attack,

the great batteries of heat-guns around the central electrostatic tower that guarded it were wheeling toward us and thundering as they shot a storm of heat-shells above the white-lit city toward us. Even as I had said, those vigilant watchers at the power-tower would have blasted our fleet from the air before we could have ever got near to the tower itself, but as it was we had struck a terrific blow at the military arsenals and the resting fleet, and had flashed upward again in time to escape the blasting guns at the city's center. But now, through the night above the vast roaring city, the batteries all around its rim were swinging their pivoted guns toward us and sending a hail of shells after us while, as all the city's great searchlights wheeled their beams madly through the air toward us, the swarming patrols all around us had recovered from their stunned astonishment and were leaping also toward us!

"One more attack!" over the uproar I was yelling into our distance-phone as we shot upward through that mad chaos of whirling beams and ships and shells. "The city's western arsenals this time—loose the other half of our bombs on them!"

Holding still to our triangular formation in that wild *mélange* of sight and sound, our ships levelled out once more, high above the city again now, and with only a scant dozen having been reached by the hail of heat-shells that had rushed after us from beneath. Then we were speeding westward over the tremendous city, high above it, scorning to stop for the swarms of patrol-cruisers that were dashing toward us. Those cruisers were rushing with suicidal fury toward us with every heat-gun detonating, but our own gunners were plying our batteries even as we dashed forward above the air-city, and on all sides of us the patrol-craft were flaring and fusing and crashing down toward the city beneath! Here high over the city, though, the shells of all the heat-guns that now were booming toward us could not reach us, and through battling ships and whirling beams and gloom of night we rushed westward over the giant air-city until in a moment more we were pausing over the western arsenals, and the western plaza where rested other massed cruisers of the great European battle-fleet. And then as I gave another order we were diving once more, down toward those buildings and cruisers!

The Second Blow

This time, though, all the colossal city beneath us was roused and roaring with fury as we shot downward, and from beneath there slanted up toward us a terrific hail of shining heat-shells from all the city's great batteries. Eastward the cruisers that had escaped our bombs there were rising and forming to attack us, while, even as we shot down, the cruisers beneath were rising and flinging themselves to one side for the moment to escape our swooping rush and bombs. But through storming shells and blinding beams we shot again on our terrific dive, until in another moment our fleet was levelling again above them and as Macklin drove our cruiser level before the rest I had pressed the remaining buttons, had sent our remaining heat-bombs whirling downward with those of all the ships about me! And then as our ships curved upward again, our terrific blow struck, the bombs were finding their mark again, were flaring and fusing with terrific heat and power into another giant mass of melting metal and awful heat there at the city's western edge.

And now, bombs gone, our cruisers were whirling upward now to escape from the great city we had struck such two awful blows, to head westward again over the Atlantic. About us a wild hail of heat-shells from the guns beneath were rushing upward and dozens of our cruisers were flaring and falling before we could gain a height again that put us beyond reach of the batteries beneath. Then we paused a moment, massing again to head westward, with only a few patrol-cruisers dashing futilely toward us from about and above us, now. Beneath us the giant air-city of Berlin lay with two white-hot craters of fusing metal glowing near its eastern and western edges amid the brilliance of its myriad lights, the great city hanging still in mid-air with the great motors in its base untouched by our two awful blows. Through its streets were rushing panic-mad crowds, and over it were rising the cruisers of its battle-fleet, striving to form and follow us as the guns thundered madly toward us and the searchlights wildly stabbed and circled.

But as we hung there for that moment, massing together again, a wild triumphant cheer was coming from all in our cruiser and all the cruisers of our mass. For we had lost but a few dozens of our ships and had all but destroyed the mighty Berlin arsenals and a thousand of the European Federation cruisers, had struck a staggering blow at our enemy. And even as

we gathered now we knew that the cruisers rising from beneath, striving to form their shattered and disorganized and stunned squadrons, would be too late to pursue us. Westward lights were gleaming in the upper air, growing larger, and we knew them to be other patrol-cruisers rushing in answer to the alarm from the city beneath, knew that even at that moment the great air-forts hanging in a chain westward would be rushing back to defend Berlin, knew that easily we could evade them and with their great chain broken could head westward at full speed over the Atlantic and win back to our own land. We had succeeded in our daring, insane plan, and our cheers were rolling out still as we began to move westward above the great, panic-roaring air-city.

"We did it!" I cried to Macklin as our cruisers leapt forward now. "We struck a blow this time that they never dreamed we had the power to strike!"

"And we'll win clear!" Macklin exclaimed, as he sent our cruiser shooting forward at the head of the others. And Hilliard, bursting up into the bridge-room from beneath, was crying, "We made it, Brant—we've destroyed their arsenals and a fourth of their fleet!"

"And now back westward!" I exclaimed as our cruiser shot ahead. "Now back—but *look there above us!*"

My words had changed suddenly into that wild cry of warning, and as the others glanced up they saw above that which had brought that cry from me. Two of the patrol-cruisers of the enemy that were dashing about us still in futile attack as we started away had drawn back and had circled upward high above us. And now, without using heat-guns and for that reason not detected by us until that last moment, they had joined together and side by side were rushing straight down upon us like a great single projectile of flying metal! Were rushing straight down toward our cruiser, that sped in front of all the mass of our cruisers, identifying it in that way as the ship of our expedition's commander, and sacrificing themselves to destroy us in a headlong crash in revenge for our bombing of the city beneath! Even as I had glimpsed them, had cried out, they were looming just above us, rushing down toward us!

Even as that wild cry had left my lips and as the others had looked up other cries had come from them, from Macklin and Hilliard. "*Over!*" I screamed

to Macklin as his hands shot to the wheel, and in the next instant he was whirling the wheel over, to send our cruiser whirling sidewise to escape that thunderbolt of twin destruction from above. But in the next moment, before it could answer to the wheel, the down-thundering ships above had crashed squarely down into our own! We reeled there with them for a single instant, three twisted wrecks of metal hanging there in mid-air in that instant, and then theirs and our own wrecked cruiser were falling, were hurtling crazily downward through the upper night toward the giant radiant circle of the great air-city miles below!

CHAPTER V

Captured

That mad whirl downward of our wrecked cruiser is now to me more of a memory of some strange and torturing dream than a memory of actual happenings. Flung sidewise and downward against the bridge-room's floor as our cruiser whirled over with that mighty crash from above, I glimpsed Macklin and Hilliard tossed about there with me, rolling over and over. The black gloom of night about us, the mass of our onrushing ships above, the colossal brilliant air-city beneath, the two wrecked cruisers that were tumbling downward with our own—all these things seemed to whirl about us like some great wheel of swift-succeeding impressions as we glimpsed them in that mad moment through the bridge-room's whirling windows.

It seemed but a single brief moment before I glimpsed the great mass of lights, the soaring towers, of the air-city beneath rushing up toward us with unearthly speed. Even as I glimpsed it another turn of the spinning ship had thrown Macklin and Hilliard over again, and this time I clutched for a hold, found one upon the cruiser's wheel. Then, with the droning of the still-operating motors and the cries of my two companions and of the crew beneath loud in my ears, I reached with a great effort toward the control of the motors, clinging to my hold with a supreme effort. My fingers found that control, but at the moment they did so I heard a last hoarse cry from Macklin, glimpsed but yards beneath us, it seemed, the smooth surface of one of the city's narrow streets, and then flung over the control, shifting all the power of the motors from our horizontal tube-propellers to our vertical ones. The next moment a blaze of light seemed all about us, there was a terrific crash, and as I was hurled back across the bridge-room by the impact, my head met the metal wall of it and consciousness left me.

When I came to it was to the realization of someone's hands endeavoring to revive me. I opened my eyes to find myself lying on a long seat of metal,

with above me the metal ceiling of a white-lit room, and with Macklin and Hilliard bending anxiously over me. I strove to speak to them, desisted as my first movement made apparent to me a painful swelling on the side of my head. And then with their helping arms behind my back I sat up, looked dazedly about me. Then, the memory of what had happened rushed suddenly back upon me and I was filled with an abrupt dismay.

For the white-lit room in which I sat, seeming an ante-room to other chambers beyond, held beside us three a half-dozen of men in the green, tight-fitting uniforms of the European Federation's forces, alike save in colour to our own black uniforms. They were ranged before us, watching us closely, and there swung at the belt of each a shining, long-barrelled heat-pistol, one of those hand-weapons that throw heat-cartridges smaller than the great heat-shells and bombs, but as destructive and deadly on a smaller scale. These six European Federation soldiers had their heat-pistols ready beneath their hands, and were contemplating us intently. And as I saw that, and glimpsed also through the open door to the right of us a great, smooth-floored plaza and immense buildings towering up into the outside night, brilliant with lights, and heard the roar of the crowds that seethed among those buildings, I remembered all that had befallen us, clutched Macklin's arm tightly.

"The cruiser fell!" I exclaimed. "I remember the crash, now—then this is Berlin, Macklin, and we're captured!"

"Captured," Macklin quietly said. "You and Hilliard and I were the only ones to survive our cruiser's crash, Brant—and we survived only because we were in the ship's bridge-room, its upmost part, when it crashed. You had been stunned, and before Hilliard and I could recover from that crash the European guards had swarmed up over the wreck and captured us, taking us here to the great central electrostatic tower."

"We three the only survivors?" I repeated. "Then—then all our crew—?"

Macklin did not answer, but as his eyes held mine I read my answer in them, and as I did so something hard seemed to form in my throat. Our crew—the hundred cheery lads that had manned my cruiser for long, and each of whom I had known by name—and all annihilated in that great crash downward which we three in the bridge-room had alone escaped. I felt Macklin's understanding grip on my shoulder, and then we were suddenly

recalled to realization of our position as a door in the ante-room's left side clicked open, another green-uniformed figure emerging from within. He spoke a brief order to our guards in the European tongue, that Latin-Teutonic combination of languages which was universal throughout the European Federation and which I myself spoke and understood to some extent. Instantly our guards motioned us to the door from which the other had emerged, and as we passed through that door before them we found ourselves in a larger and circular room, white-lit like the first.

It was, I saw instantly, the central control-room of the great power-tower, of the whole great air-city of Berlin. Like the similar control-room in the power-tower of New York it held on its walls panel upon panel of dials and gleaming-knobbed switches, while at the center of the room were also six great controls that directed the great air-city's movements through the air in any direction, and the single power or speed-control. Beside these was another great raised table-map, this one mounted upon a solid block of metal, with upon it the red circles of the world's air-cities. And beside that map there sat now a dozen or more men in the same green uniform as our guards, though with metal wing-like insignia upon their sleeves. They were, I knew without asking, the highest Air Chiefs and officers of all the European Federation, gathered here in the control-room of that Federation's capital city.

The Captors' Threat

For a moment we three faced them in silence, our guards watchful still behind us, and then the center-most of the seated figures, a swarthy, black-haired officer with black, probing eyes, whose five metal wing-insignia marked him as the First Air Chief of the European forces, spoke to us, in our own tongue.

"You are Captain, First Officer and Second Officer of the American Federation cruiser which crashed in our streets just as the main body of your ships escaped," he said, and even at the words my heart raced with sudden gladness. Our ships had escaped safely back over the Atlantic, then, as I had known they would! "——and we desire to know," the European

First Air Chief was continuing, "just what forces remain to the Americans and which engaged in this attack."

I faced him in utter silence, my own eyes meeting his probing black ones calmly, and at my silence I saw a contraction of the muscles about those eyes, a sudden flush beneath his swarthy skin.

"I think it would be best for you to answer," he said quietly, "nor need you think that silence will help your countrymen in any way. For though your cruisers struck a great blow at us here in Berlin this night, though word has reached me that as great a one was struck by other American ships at Peking, these are but two of the two hundred great air-cities of our two Federations, but a fraction of our great forces. And we know that your fleets lost many ships in the battles of yesterday despite their victories, and desire to know what forces are left them."

Still in stony silence I stood, my eyes meeting squarely the eyes of the men before me, while beside me Macklin and Hilliard stood in the same stiff silence. I saw the European Commander's flush of anger deepen, saw him half-rise with hand clenched to hurl an order at our guards, and then he had relaxed back into his seat, was smiling grimly.

"A most unwise course to follow, Captain, you may believe me. I take it that your officers are as mule-headed? Well, there is no immediate hurry and a few days of consideration, of meditation, may change your minds. As a subject for your meditations, you may take my promise to you that unless you become more communicative at the end of the fortnight I give you, we shall be forced to use somewhat unpleasant procedures with you. An earnest consideration of that fact will, I think, change your viewpoint somewhat."

He turned, snapped an order in his own tongue to the captain of the guards behind us. "A cell in the one hundredth story for these three—put them with the other American, and if after a fortnight they're still stubborn, we'll deal with all four."

Immediately our guards had marched us back to the door through which we had entered, and across the ante-room beyond through another door and into a short, broad hall along the sides of which rested the great tower's lift-cage. We were ordered into one of the cages, our guards holding their heat-pistols full upon us now, and then as a stud was pressed and the motors'

power was turned through the cage's powerful vertical tube-propellers, those tube-propellers drove us up with a thin whistling of air up through the narrow shaft the cages moved in, up until in a moment more we had stopped and were emerging into a similar hall on the great tower's hundredth floor. From that hall we moved into a short corridor that ran the width of the great tower, which at this height was but a hundred or more feet in diameter, its slender pinnacle tapering as it rose to its tip, while much of that pinnacle's space was occupied by the great connections which carried the city's electrical power down from the mighty tower's tip.

Along that corridor we went, one lined with solid metal doors on either side, and finally were halted before one of those doors. Then one of our guards drew from a pocket a small instrument resembling an electric torch, from which he flashed a tiny beam into a transparent-fronted little opening in the wall beside the door. At once there came a clicking of locks, and the door swung open, its locks unbolted by the beam of light or force, rather, whose vibratory rate was exactly tuned to affect a delicate receiver tuned to the same frequency, set in the wall and controlling the lock. These vibration-locks, indeed, had long ago replaced the old, clumsy keys, and were far safer in that they responded only to one certain frequency vibration out of the millions possible, and thus could be opened only by one who knew the correct frequency. Now, as the door swung open, our guards pointed inside with their heat-pistols and perforce we stepped within, the door snapping shut behind us.

We found ourselves in a small, metal-walled cell some ten feet in length and half that in width, furnished with but a few metal bunk-racks swung from the walls. At its farther end from us was the only opening beside the door, a small square window that was quite open and unbarred, and that looked out over all the colossal mass of the great air-city of Berlin, a giant field of blazing lights stretching far around and beneath the great tower in which we were prisoned. Then, as we gazed about the little cell with our eyes becoming accustomed to its lack of light, we made out suddenly a figure standing near its window, a dark, erect figure who seemed watching us for the moment and who then was striding across the cell toward us.

"Brant!" he exclaimed, as his eyes made out our faces through the dusk. "Brant—and you were with the ships that attacked the city but now—you were captured in some way!"

But now my own eyes had penetrated the dusk enough to recognize the features of the man who was gripping my arms, the keen, daredevil countenance that I remembered at once.

"Connell!" I cried. "You prisoned here! Then you're the other American the European First Air Chief ordered us prisoned with. But I had thought you dead!"

"Dead I might be as well as here," said Connell, suddenly somber. "For four weeks I have been here, Brant—for weeks before the beginning of this war. And now that this war has begun I, who alone might save our American Federation from annihilation in it, am prisoned here with only death awaiting me, and that in a few days."

I stared at him, astonished. Connell had been one of the cruiser captains of the American Federation forces for several years, and had been a friend of my own in those years. A year before he had withdrawn from active duty, no one knew to where, and finally, but a few weeks before the breaking forth of this war, our First Air Chief had told us in answer to our queries that Connell had been sent upon a special mission, but that since he had not reported for several weeks he had undoubtedly met death in the course of it. To meet him here, in the heart of Berlin and prisoned with ourselves, astounded me, and the more so since from his first words we understood that he had been confined thus for weeks even before war had burst upon us. But now, motioning us to seats on the bunk-racks beside us, Connell was questioning us eagerly as to the course of the combat between the great Federations so far, and his eyes shone when we described to him that terrific battle over and in the Atlantic that we had fought but a day before, and that daring attack on Berlin that he had himself witnessed from his window.

"I saw the European Federation's fleet massing and sailing westward yesterday," he said, "and knew it was launching its great attack, knew when it returned disorganized and shattered that the American fleet had beaten back that attack. But I did not expect this attack you made on Berlin tonight, and was as astounded as all in the city when you swooped down with your great bombs. A great blow, Brant—a great and successful blow against the whole European Federation, yet such a blow alone cannot halt the menace which it and the Asiatic Federation are preparing to loose upon

our own nation. Such a blow, nor a hundred such blows, would avail but little in the end against the stupendous plans and forces that are preparing and massing even now to roll out upon the American Federation in an avalanche of doom!"

A Strange Tale

He paused, and in the dusky cell Macklin and Hilliard and I sat as silent as himself, gazing toward him in sudden startled surprise. From far out over the great air-city about us came the droning of rushing ships and the dim roar of voices from beneath. But Connell was speaking again—

"You, nor anyone else, knew where I went when I left active service in our fleet, none but the first Air Chief, who sent me. That was a year ago, and he told me then that it was evident that the European and Asiatic Federations were preparing to attack us, and that rumors had been heard of some mighty new weapon or plan with which, if their ordinary forces failed, they would completely crush us. Hundreds of agents, said the First Air Chief, were being sent to the European and Asiatic air-cities to try to learn the nature of this new weapon, and I was one of those to be sent to Berlin, as I knew the European tongue thoroughly. I was to go in disguise, was to endeavor to work myself into the European Federation fleet, and was then to risk everything in an effort to find out what this great new plan or weapon was. And so in disguise, a year ago, I came here.

"Eight months it took me to work my way into the European fleet, eight months in which I was chiefly occupied in establishing my new false identity as a European citizen. Then I enlisted in the fleet, entering the motor-section. Of course, as a cruiser-captain in our own fleet, all types of motors were perfectly familiar to me, and I had no difficulty in swiftly rising through various promotions to the status of under-officer in one of the European cruisers. Then came at last the opportunity for which I had waited for months, and which I had begun to despair of ever occurring. I was ordered to report back from my cruiser to the First Air Chief's headquarters here in Berlin, and when I did report I was questioned by a board of a half-dozen European officers on my knowledge of motors and tube-propellers. It

must have seemed to them that I had unusual ability and knowledge for a mere under-officer, for they informed me that I had proved satisfactory and that I had been selected to form one of the workers on a great new work that was being carried out secretly, and ordered me to report to a certain compartment in the great air-city's base.

"I reported there, eager now as I sensed myself on the trail of that which I sought, and found that there were whole vast compartments in the city's great base in which only selected men and certain high officers of the European fleet were permitted to venture. These were the compartments in which were placed the giant tube-propellers which are set horizontally in the great air-city's base, and which when the power of its great motors is turned into them move the city in any desired direction. Every air-city in the world has, as you know, these great tube-propellers that move it about. But as you know too, so much of the motors' power must be used in the life of the city, that the horizontal tube-propeller can only move the great cities through the air at an extremely slow rate of speed. It is a predicament which cannot be altered, either, by adding more motors, since to add them you must add to the city's size, and so the problem remains the same.

"But now, as I found when I first entered those compartments, these European Federation officers and inventors had solved that problem! They had devised a way that would enable them to send their gigantic air-cities rushing through the air at almost the speed of a cruiser itself! They had done this by devising a wholly new form of horizontal tube-propeller capable of infinitely greater tractive effect on the air and rotating at a much higher rate of speed. Thus the great air cities, miles across and with all their towers upon them, could rush through the air at hundreds of miles an hour, needing only to use their vertical tubes when they were hovering motionless in mid-air or were moving very slowly.

"And *this* was the great weapon, the great plan, of the European and Asiatic Federations! For I saw at once that it was a great weapon indeed, a terrific weapon which would enable them to annihilate all the air-cities and peoples of our own nation. You see what it meant? It meant that they could gather together all their scores of giant air-cities, outnumbering our own one hundred cities by two to one, and could rush over the oceans at awful speed toward our American air-cities, could fall upon them with all the giant batteries of heat-guns with which each colossal city is equipped, like our

own. And because our own would not be able to move at that tremendous speed, because our own air-cities could only move at a comparatively creeping rate through the air, they would be able to mass their outnumbering forces around our own cities and blast them from the air, annihilating them and all the millions of our people inside them, sending them hurtling to earth in titanic fusing wrecks!

"To rush forth to battle, to the annihilation of our own cities, in their great air-cities! To send those gigantic cities of the air, Berlin and Peking and Tokio and all the scores of others of the two great Federations, thundering through the air to battle, each with its masses of towers on it. They have made provision for all people who are not entirely engaged in battle, to descend to the earth and remain there in specially constructed buildings. This will help also to reduce the weight of the cities. That was their great plan, their great weapon, and I knew that with it, even as they said, they could burst forth and annihilate our own air-cities. But, holding still to my work there in the lower compartments, I strove to penetrate the heart of the secret, the design of the great new horizontal tube-propellers which were to accomplish this, to send the mighty cities rushing through the air at such immense speed. Each of the great air-cities of the European and Asiatic Federations, as I learned, was being secretly equipped with these new tube-propellers, and I knew that if I could learn their secret, could take that secret back with me, our own American air-cities could be equipped with the new tubes likewise and could meet the attacking cities at equal speed, on equal terms, even though outnumbered.

The Great Danger

"So I endeavored in every way to penetrate the secret of the new tubes, to ascertain their construction, which was jealously guarded by the European and Asiatic Air Chiefs. And at last, hardly a month ago, I did that, was able to make my way from my own work to one of the great tube-propellers which was being installed in another compartment, and by taking a place among those working on it was able to learn the details of its construction. That construction was simple enough, I found, amounting in fact to hardly

more than a use of many smaller tubes within the main tube-propeller, smaller tubes which drew air from different directions upward and ahead, and thus by their shaping and construction were able to fling a great air-city supported by them onward through the air at that tremendous speed. I had learned the great secret for which hundreds of our agents had sought, and needed only to escape with that secret.

"I needed only to escape, to race back to my own land, and knew that it would take our own engineers but a very short time to fit our own cities with similar speed-tubes, since though the European and Asiatic forces had been working with them for months that work so far had been mostly experimentation. But it was then, when I tried to escape, that my luck came abruptly to an end. For I was captured by the fleet-officers here in Berlin as I was on the very point of leaving, captured when the false identity which I had established at such pains was upset at the last moment through the detection of one of the documents I had forged. I was captured, and knowing that I had within my brain that great secret of theirs which would make their air-cities resistless, they would never, I knew, release me. They took me at once before their commander, the First Air Chief of the European fleet, and then by him and by a number of the Asiatic Air Chiefs also I was questioned exhaustively.

"They wanted most to know what other American agents like myself were hidden within their air-cities. They knew that those agents or the greater part of them were known to me, and they knew that if I described or named them they would be able to catch them all and thus prevent the possibility of another spy learning their great secret as I had done. I refused utterly, though, to give them the information they wished, to reveal to them my fellow-agents in the various cities. At last they saw, after days of questioning and half-torture, that they could not as yet wring from me that information, so confined me here in a cell high in the central tower with the information that only death awaited me within days unless I acceded to their demands. And, confined here, I saw from the window that the whole European Federation fleet had begun to mass here at the air-city of Berlin, quietly and unobtrusively, and guessed then that they meant to loose their attack upon the American Federation.

"The great tubes that were to move their cities through the air at such terrific speed were not yet finished, but they did not wait for these,

launching out their great fleet of cruisers which with the Asiatic fleet outnumbered the American ships by two to one and should be able to overwhelm them, they thought. I think that their reason for starting that attack so soon, before their greater preparations were completely finished, was that they feared lest another spy like myself might discover their great secret and escape with it. So they let loose their fleets upon the American Federation to begin the war and forestall that contingency by beating down the American forces in a first tremendous attack. If that first great attack failed, they could swiftly complete the preparations that would make their air-cities of such immense speed and power, and then could launch all those air-cities upon the American ones in a second attack that nothing could resist.

"And even now, despite that daring and deadly attack which your ships made here upon Berlin tonight, and upon Peking, as you say, the great preparations of the European and Asiatic Federations are going swiftly on, and soon now those preparations will be completed and their great air-cities will be able to whirl through the air at that tremendous speed. And then will come the end, for our American Federation. The two hundred air-cities of the European and Asiatic Federations will flash upon our own nation from east and west, with all their millions of people and giant batteries of heat-guns, and will send our own slow-moving air-cities crashing to earth, will send all the scores of cities and all the millions of people of the American Federation into destruction and death!"

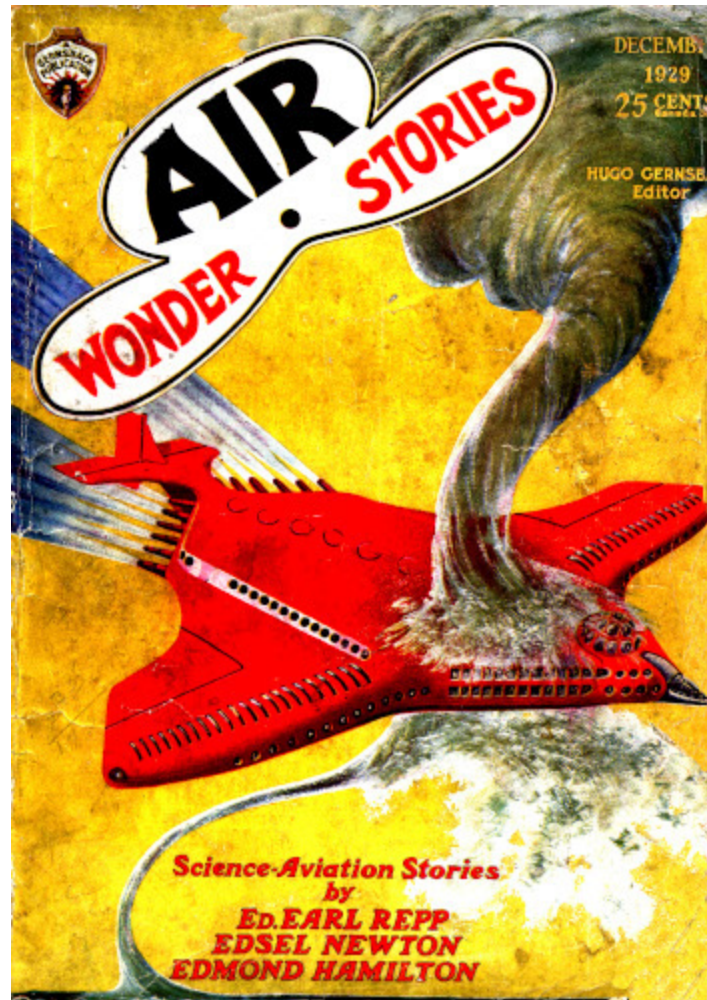
"Destruction and death!" Connell's voice seemed echoing still about us there in the silence when he had ceased, seemed beating like great drum-notes of doom in our ears. Macklin—Hilliard—they sat beside me in the dark cell as silent as myself. And in that moment we heard again, from outside and far beneath, the great throbbing roar of the life of all the mighty air-city about us, the humming rush of cruisers to and fro above it and the dull mingled voices of its great crowds, coming dimly up to our silent little cell high in the mighty electrostatic tower. Then suddenly I had risen to my feet.

"Destruction and death—but there must be some way in which we can prevent it!" I cried.

"What way is there?" Connell's tone was low, hopeless. "We only know what looms above our nation, know that these preparations are coming to their end, that these air-cities plan to rush upon our own. We cannot halt the preparations that are going on in every air-city of the two great enemy Federations."

"But if we could warn our own!" I said. "If we could get what you have learned back to the American Federation—could install in all our own air-cities similar new tube-propellers—then our cities could at least meet the attack of the enemy cities with equal speed and power."

"But how to get back?" asked Connell. "How to escape from here? It could be done, if we could escape, for the new tube-propellers could be put in our own air-cities swiftly enough, yet to escape is impossible. I have been here days, weeks, Brant, with the one thought of escape uppermost, but the thing is hopeless."



I strode to the square little window, looked forth from it. It was quite open and unbarred, and large enough too to allow one to pass through it, yet as I projected my head from it and gazed up and downward in the darkness I saw that there was no need of bars across it. For the little window was set directly in the sheer, towering side of the mighty power-tower's pinnacle. Far up above our level soared that tremendous tapering tower, so far that the tip seemed among the stars above, while far below, a thousand feet at least, lay the smooth metal of the great plaza. And though there were other windows below and above us, each was separated a full ten feet or more from the other, and, as we knew, to merely escape from our cell into another level of the great tower would avail us nothing, since to gain the plaza

outside we would need to pass through the tower's lower levels thronged always with armed guards. It seemed, indeed, that as Connell had said there was no hope of escape for us, the door being solid, thick metal, and as I turned back toward the other three something of Connell's own hopelessness had taken root within my heart.

And that hopelessness grew within me in the hours that followed. For when day came and illumined with brilliant light all the giant air-city that stretched far around us it seemed only to emphasize the utter helplessness of our position. Far beneath on the great plaza lay many cruisers, and could we win to one of them we might well make a break at top speed across the Atlantic, since so simple in design and so unvarying in their exhaustless power-supply are modern air-cruisers that one man alone at their bridge-room controls could operate them. Yet to win down to those cruisers, down to the great plaza's surface—that seemed impossible. And so as that day waned, and night swept over the great floating mass of the towers of Berlin, to be followed by day again, my despair was waxing ever stronger, deeper.

For during those days we could see plainly from our window the great preparations going on still in the air-city about us. Already throngs of workers had cleared away the twisted and fused wreckage that had been made by the attack of our ships, and new masses of supplies were pouring into Berlin in shipload after shipload from all the air-cities of the European Federation, to replace those we had destroyed in their great arsenals. The air seemed filled, indeed, with great freight-carriers and official cruisers arriving and departing. And beneath all this great surface activity and preparation, we knew, down in the great tube-propeller compartments of the air-city's mighty base, other and greater preparations were going on, other and different tube-propellers were replacing the city's tubes, and swiftly the time was approaching when all the city would be able to rush meteor-like through the air.

It was that knowledge that made our despair most deep. For though there was now a lull, apparently, in the great war's course, the European and Asiatic forces preparing for their final giant blow, and the Americans gathering their own forces apprehensively to resist the next attack, we knew that it was but the lull before the final terrible storm that was to settle the fate of earth's three mighty nations. And we knew, too, that it was the fate of our own American Federation that would be sealed in that gigantic attack,

unless Connell could make his way soon to our land with his great secret. And that he could not do so, that he could not even escape from the little cell in which we were prisoned, was all so clear to us that almost I wished that death had come to me in the cruiser's crash to spare me the torture of mind that I and all of us were now undergoing.

It was a torture accentuated, I think, by the complete emptiness and eventlessness of those hours and days. Save for what we could see from our high window upon the city around us, we were as cut off from the world as though upon the moon. Twice each day, at dawn and at dusk, our door was opened by the guards that brought our food, that food being as in our own air-cities the paste-like synthetic compounds of artificial proteins and fats and carbohydrates which had decades before replaced the old natural foods. But though our door was thus flung open twice each day, there was no hope of escape for us in that fact. For the two guards who brought our food in to us carried their heat-pistols always in one hand, and always, night and day, there watched in the corridor outside a full score of similarly armed guards by whom one could not hope to pass living toward the cage-lifts. It seemed indeed, as Connell had said, that weeks of frenzied meditation could never disclose any plausible plan of escape, and so I lapsed with him into a state of half-lassitude that had been induced by our utter despair.

And so days passed. Not even the prospect of our own deaths which I knew to be looming before us, was sufficient to rouse me from that lassitude, not even the fact that at the end of that fortnight, as I had guessed, the great attack of the air-cities was to be launched upon the American Federation, and that it was for that reason that our captors had given us that time.

Connell, Macklin, myself—we three had faced in our time perils and risks enough, but so overwhelming was the doom that hung over us and over our nation now that it stunned us, held us in stupefied despair. But one of us there was that was not so stunned, and that was Hilliard, my young second officer. His eager, restive nature, chafing at our imprisonment and at the thing that was looming for our land, resisted stubbornly the deep hopelessness that had settled upon the rest of us, and hour after hour he spent in pacing about the little cell, or in striving to devise some means for escaping from it. And at last, upon the fourth day of our imprisonment there in the tower, he turned suddenly toward us with an eager cry upon his lips.

CHAPTER VI

A Daring Venture

"I have it!" he exclaimed. "A way that two of us can win free with—and maybe all! A chance out of millions, with death at the end of it, maybe, but a chance to get Connell and his secret back!"

"But how?" I questioned. "How can two of us, even, get clear of this cell?"

"The guards!" he exclaimed excitedly. "The two guards that bring our food each dawn and dusk—if we can overpower them—"

"It's useless, Hilliard," I said. "Even if we did overpower them we could not pass the score of other guards in the corridor outside, or the scores of others on its lower floors."

"But listen," he appealed, and then, as he went on to detail to us the plan that he had devised, I felt some slight measure of hope rising within me, saw that dawning hope reflected in the faces of Connell and Macklin.

"It *is* a chance!" he exclaimed. "And if we can do it, if I can get back to our own land, it means a chance still for the American Federation! For the European and Asiatic Federations won't be starting their attack for another ten days or more, their preparations not finished till then, and in that time by bending every effort toward it our own engineers could put the new tube-propellers in all our American air-cities! Could make those cities able to meet the enemy cities, attack when it comes!"

So, with that foremost in mind, we swiftly decided that upon that very evening, when our guards brought our food at dusk, we would put the plan into operation, would stake everything upon it. For even if but two of us could escape by it, if Connell could be one of those two, and could get back across the Atlantic with his tremendous secret, it meant a fighting-chance for our Federation. And with that in mind the rest of us were willing to take

all chances, to dare all risks. Risky enough the thing would be, we knew, all depending upon what occurred in one moment of rushing action, and numberless were the features of it that might go entirely wrong and ruin us. But we steeled ourselves with the thought of what would become of our Federation if we failed. And so ready and tense with resolution we waited for the coming of dusk.

To me, through the remaining hours of that day, it seemed that never had the sun sunk westward so slowly. From our window we could see all the activities that went on in the great air-city about us, could see far across all this great mass of towers and streets and thronging crowds which hung here miles in the air above the earth, and could see those activities lessening somewhat as the long shadows of sunset fell across narrow streets and smooth plazas. In the great plaza beneath, at the foot of that electrostatic tower in which we were imprisoned, there rested always a number of great cruisers of the European fleet, reporting to the First Air Chief there in the tower's base, and now with the approach of night other cruisers from the swarming masses above the city were slanting down to rest upon the plaza. And these cruisers we watched with intent gaze as the sunset's light declined.

Outside in the corridor we could hear the occasional movement of feet as the score of guards there moved about now and then, but heard not the approaching feet of the two that brought our food. What if they were not to come upon this evening? Or what if more than two, or less than two, were to come? Either contingency would be equally ruinous to our plan, and with the passing of each moment we sat in an increasing agony of expectation, Connell's eyes burning, Macklin as imperturbable as ever, Hilliard eager and tense. Then at last, when the shadows of dusk were falling across the great city of the air outside, were deepening in our little cell, we heard voices outside, the greetings of our guards in the corridor, and then a moment later the solid metal door of our cell had clicked open. Then into the cell stepped our two usual guards with our food, their heat-pistols ready as always in their right hands.

The eyes of one warily upon us, the other took both of the metal containers of synthetic food and reached to place them, as usual, upon the lower bunk-rack, at the room's right hand side. Macklin was lying upon that bunk-rack having stretched himself out as though sleeping. The rest of us were

lounging at the cell's other side, the second guard's heat-pistol watchfully upon us while the first reached toward the bunk-rack to place the metal food-containers beside the supposedly-sleeping Macklin's head. But as he placed them there, as he began to turn away from it, Macklin's hands shot suddenly behind his head as he lay there and grasped the arm of the first guard in a single movement, jerking him toward the bunk-rack! Like lightning the second guard turned with his pistol toward that sudden movement and as he did so, forgetful for the instant of the rest of us, we three had leaped upon him! And then as Hilliard and I bore the second guard to the floor, wrenching the pistol from him, Macklin and Connell had jammed the first one against the cell's corner, with hand upon his mouth, and had him equally powerless!

The whole swift scene of action had taken but a flashing moment to carry out, and so lightning-like had been our movements, so careful above all had we been to gag with our hands the two guards as we grasped them, that no single sound save for a few low-choking gasps had come from them. And then, while with hands over their mouths and with all the strength of our muscles Hilliard and I held the two guards, Macklin and Connell were swiftly stripping their tight-jacketed green uniforms from them.

A moment more, and Connell and Macklin were swiftly doffing their own black uniforms of the American Federation and donning the green ones. Now came a restless movement of feet in the corridor without, and with the speed of utter necessity we took the two discarded black uniforms and forced them upon the two guards, holding them still voiceless and powerless. Then, that done, the two guards were as like in their black uniforms as Hilliard and myself; with Connell and Macklin, the latter having been chosen because of the similarity of his appearance to one of the guards, wearing now the guards' green uniforms. And now the very climax of our endeavor was come, that moment upon which depended all, for now, suddenly removing our hands from the lips of the two guards we held, we added our own sudden cries to theirs, and at the same moment Connell and Macklin, in their green uniforms, were engaging in a mock struggle with us and with our guards, whose din seemed terrific there in the quiet upper levels of the great tower!

Instantly as those cries arose there was a rush of feet outside and the score of guards there poured down the corridor and through the cell's door, to see

four of us in black uniforms struggling apparently with two green-uniformed guards, who were in reality Connell and Macklin. It was the moment upon which all rested, and in that moment the score of guards acted as Hilliard had foreseen, gazing not in that wild moment of frenzied action at faces but at uniforms, seeing only in that first moment in the dusky cell four black-garbed men struggling with two green-garbed ones they had seen enter it but moments before. And, seeing this, they rushed upon us four black-uniformed ones, Hilliard and myself and the guards whom in the guise of aiding in the struggle we had held, and began to beat us back against the cell's end, totally forgetting in that moment the two green-uniformed men they were succoring! And in that moment, as they pressed us back, Connell and Macklin were stealing swiftly out of the cell, and down the corridor toward the hall of the cage-lifts!

For but a moment more did we struggle with our outnumbering opponents there, and then as they gripped and held the four of us helpless and jerked our heads up wild cries came from them as they saw the faces of us, saw that two of us were their own guards! Then the next moment, needing no other explanation of what had happened, they turned with their own black-garbed fellows, rushed out of the cell, slamming shut its door behind them, down the corridor and toward the cage-lifts after Connell and Macklin! We heard those cries re-echoed swiftly over all the mighty building, heard rushing feet and repeated calls on the levels of the great tower above and beneath us. And then, as we flung ourselves to our window, gazed downward, we saw in the next moment two green-garbed tiny figures issuing from the giant electrostatic tower's base far beneath us, running across the great plaza toward the nearest of the cruisers. They were Connell and Macklin!

I cried out with Hilliard hoarse words of encouragement, forgetful that none could hear from our terrific height, saw Connell and Macklin rushing up the slender gangplank in the cruiser's side and through its open door, saw that door slam shut behind them, just as from the base of the tower there poured out after them a flood of pursuing green-garbed figures. Those pursuers were raising their heat-pistols, and a hail of shining heat-cartridges were flying through the air toward the cruiser in the next moment, but as they shot toward it there came faintly up to us the droning of the ship's great motors and then with all the power of those motors in its vertical lifting

tube-propellers it was rising upward, was lifting smoothly and swiftly upward toward us! And at the same moment the green-uniformed crowd of guards beneath were rushing across the plaza toward the other resting cruisers, were rushing within them to soar up after Connell and Macklin!

Up toward us shot the lifting cruiser of our two friends, though, up toward our little window, for such had been our plan by which two, escaping, might rescue the remaining two. But as it shot upward I looked down, saw beneath the scores of cruisers on the plaza rising now after it, heard through all the great tower and for far around it a great roar of rising shouts as the escape was discovered, saw the giant gleaming muzzles of the great batteries of heat-guns around the plaza turning swiftly upward on their pivots toward the rising cruiser of Connell and Macklin! And so the next moment as that cruiser shot up toward us, as I made out Macklin plain in its transparent-windowed bridge-room, driving it up toward us, I flung an arm outward from the tower to him, shouting in frenzied appeal.

"Back, Macklin!" I cried, with Hilliard crying too beside me. "No time to get us—*back home with Connell, for God's sake!*"

He saw my frenzied gesture westward, caught the meaning of my wild warning shout as the guns beneath swung toward him and the cruisers below rushed up, and I saw him hang there for a fraction of a moment irresolute, hesitating. Then the next moment, just as there came a swift-spreading thunder of detonations from the great heat-guns around the plaza he had whirled the wheel over and sent the great cruiser rushing away from the tower, sent it rushing westward through the dusk above the great air-city's gathered lights. In the next instant there shot through the air where it had been the shining heat-shells from beneath! And then as Macklin's cruiser rushed comet-like onward through the dusk the great heat-guns beneath were turning again toward it.

I cried out hoarsely as they thundered again, but with a whirl sidewise Macklin and Connell had evaded the rushing shells and were hurtling on. Now over all the great air-city, over all the mighty mass of Berlin was spreading a roar of alarm, and now the cruisers that had rushed up in pursuit were rocketing westward after that single fleeing one, the batteries beneath us holding their fire lest they strike their own pursuing ships. With our hearts pounding Hilliard and I saw that single little cruiser leap on, saw it

shooting through the dusk until its gleaming shape was now far away from the great air-city, racing westward! Swiftly, though, the numberless pursuing cruisers were converging upon it, and then, as we strained our eyes to see the flying gleaming craft, there came a greater thundering of guns as all the suddenly-alarmed batteries at the air-city's westward edge loosed their shells upon the fleeing cruiser! That cruiser seemed to halt for a moment unaccountably, there was a great blinding flare that could be made only by heat-shells striking, and then the cruiser, the cruiser that held Macklin and Connell and all the American Federation's fate, was reeling blindly downward and out of sight, whirling lifelessly downward toward the earth far below!

CHAPTER VII

The Great Movement Starts

Stunned and stupefied, Hilliard and I gazed out in that moment from our window, out through the dusk above the air-city to where the cruiser of our two friends had plunged to death. I think now that for those first few moments neither of us was able completely to comprehend what had happened, to comprehend what malign fate it was that had sent our friends down to death there as they seemed making their escape. Staring forth blankly, we saw the cruisers that had been pursuing them, that had been overtaking them, turning back now toward the air-city, heard a cheer rolling across that city as the crowds in its streets witnessed the destruction of the fleeing craft, the flare of the shells that had destroyed it. That great roaring cheer from beneath penetrated at last into my brain with realization of what had happened.

"Macklin—Connell—" I whispered. "Macklin and Connell—gone—and the last chance to warn our Federation gone—"

Hilliard's eyes were bright with unshed tears. "Our last chance," he said.

Looking back, I think now that it was not the passing of our one chance for freedom, nor the passing even of our one chance to carry Connell's great secret homeward, that weighed upon us most in the following time. It was the swift passing of our two friends, of Macklin especially, who for long had formed with Hilliard and myself the trio that commanded my cruiser, that stabbed us most in those first following hours and days. Prisoned there as before, but two of us now where there had been four, we waited now in a certain heedlessness for the doom that we knew awaited us and our Federation. The wild break for freedom that two of us had made and that had ended in those two's destruction, had apparently not changed the plans of the European First Air Chief in regard to us, and we knew that at the end

of the designated fortnight, less than ten days hence now, we must either reveal all our knowledge of the American forces, which we could not do, or suffer death.

We knew, too, that even as Connell had guessed, it was at the end of that fortnight, ten days hence, that the European and Asiatic Federations planned to launch their final gigantic attack of air-cities, since it was evident that they wished to gain their information from us only to use it immediately in their attack. For now below, in the city's base-compartments, the great new tube-propellers that were to whirl it through the air at such terrific speed were being completed, we knew, as in all the two hundred air-cities of the European and Asiatic Federations. The long months of experimentation over, it needed but weeks or days to rush those new tube-propellers into place. And had Connell escaped with his secret it might well have been, I thought, that even in the ten days left the new-type tubes could have been swiftly manufactured by thousands and placed in all our own American air-cities.

If Connell had escaped! But Connell had not escaped, Connell had plunged to death with Macklin, amid the flaring heat-shells. Prisoned there in our little cell, Hilliard and I despite that ever-approaching doom almost paid no attention whatever to all outside and about us, brooding there in silence hour upon hour as night followed day and day night. We had not, even, the slightest further thoughts of escape, although such thoughts would have been hopeless, for now our door was never opened save by the full score of armed guards outside. So, losing all thought and all hope of freedom, we sat on in our little prison high in the mighty tower, dead to all the unceasing rush of preparations and gathering of cruisers in the city about us.

But at last, upon the eighth day after the break of Connell and Macklin, and the second day before our approaching doom, there came an event which roused us suddenly from that renewed apathy into which we had fallen. For days we had noticed that the crowds in the streets were proving fewer and fewer, the only people now remaining being groups of green-uniformed officials unceasingly moving in and out of the headquarters there. There was finally made clear to us the reason of their activities. For, as we gazed forth from our window on the afternoon of that day, we seemed to sense a certain air of anticipation in the people that remained. They swarmed forth into the great air-city's streets; we heard in a moment more a strange great

hissing from far below us, all around the city's base and edge; and then were aware that with that hissing sound and with a great tremor of power that beat through all its colossal metal mass, the great air-city was moving! Was moving not slowly and majestically as air-cities commonly move, but was leaping forward through the air with sudden tremendous speed. We knew now that most of the city's population had been removed to the ground and the movement toward the west had started.

Now came excited roars from the crowds beneath, as the giant mass that was Berlin leapt forward, and now as Hilliard and I leaned from our window with an excitement almost as great we caught our breaths. For we could see now, from the cloud-masses that lay beneath in the distance, that the great air-city was cleaving the air at a speed that was rapidly mounting to over a hundred miles an hour. Terrific winds were whirling all about our power-tower, as it shot through the atmosphere, and those same winds sweeping with titanic force through the city's streets and about its towers forced the crowds in those streets swiftly within the shelter of the structures. And still at ever-mounting speed, the hissing of power and the tremendous roar of winds increasing still, the mighty air-city was whirling on, its soaring towers of metal swaying back beneath the awful winds of their progress, whipping through high cloud-banks and out into clear air again, giving us flashing glimpses from our own wind-swept window of the ground far outward and beneath flashing back at immense speed as we shot onward, as all the colossal city sped on, at a velocity that I knew by then must be over a hundred and fifty miles an hour!

A colossal city, speeding through earth's atmosphere! Awed, despite ourselves, Hilliard and I clung at our window there as with all else in the city we sped on. A colossal city five full miles in its diameter, with all its works and streets and giant batteries of heat-guns, and rushing above earth at a velocity seeming almost unattainable! And even as we watched, we felt the great city slanting upward with the same terrific speed, climbing swiftly upward until the air about us was all but freezing and then diving down toward earth once more on a long, gliding swoop! Then it had turned in mid-air, was flashing back over its course, was going through maneuver after maneuver until at last the great hissing from its base ceased, and it hung at its former height above the earth once more, the crowds in its towers surging forth now to renew their excited shouts.

Last Preparations

Hilliard and I gazed for a long moment at each other. "The tube-propellers they were putting in—finished—," he said slowly, "And Berlin ready now for the great attack—"

"And all the other European Federation cities," I said, "and all those of the Asiatic Federation—all must be nearly completed now, their new tube-propellers installed also. And in two more days——"

In two more days! It was the thought that beat hammer-like in my brain and in Hilliard's in those hours that followed, those hours that were now closing down, one by one, upon the doom of ourselves and of all our nation. Two more days! Two more days at the end of which would have ended the fortnight of our imprisonment, when would come for us the death that had loomed larger and larger during each of those passing days. Two more days at the end of which the great air-cities of the European and Asiatic Federations would rush like whirlwinds over the oceans toward our own slow-moving and helpless cities, to beat them down with all the thunder of their giant batteries. Two more days!—and at the end of them for us and for all the great air-cities and all the millions of the American Federation, doom!

It is not to be wondered at, therefore, that in the following hours, Hilliard and I felt close about us the intense despair that ever since the ill-fated attempt of Connell and Macklin had surrounded us. Through all that night following the first demonstration of the immense speed of the air-city, we sat awake, listening to the great shouts of triumph and exultation that came dimly up to us from the crowds that remained in the streets far beneath. The European Federation, we knew, already felt the glow of imminent victory that this new speed of their great air-cities would give them, and were exultant at the chance to annihilate completely the hated American Federation. And, to accomplish that, the very last great preparations were going on now in every part of the air-city.

Great loads of shining heat-shells were being transferred from the stores that had been brought to Berlin, to the giant batteries of heat-guns around

the city's edge and its central plaza around the electrostatic tower. The cruisers of the European battle-fleet, still some two thousand in number, were resting on all the landing plazas, and were being cared for and inspected by hordes of green-uniformed attendants. All other air-craft were lowered into the great city's base-hangars to be out of the way during the oncoming combat. By a stroke of genius on the part of Berlin's commander, the power of the great air-forts had been added to that of the city itself, by simply placing the air-forts here and there on unused landing plazas, where they formed in effect great armored gun-turrets on the city's surface. And, finally, the mighty city's speed and power to maneuver had been tested rigorously. With all its peoples inside its metal towers, it was shot at terrific speed low and high above the earth; turning and dipping and rising at that awful velocity like a flashing airplane rather than a gigantic city of the magnitude it really had.

Through all the hours of that night, and the next day (the thirteenth of our imprisonment) those great preparations, that unceasing rush of excited activity, continued. Night came, and still the last preparations were to be made; magazines were being filled, and green-uniformed figures were swarming in countless numbers in the streets; going about their maneuvers; battle-cruisers were moving ceaselessly across the sky. During the hours of that night, as Hilliard and I sat silent there, high above all the tremendous turmoil of the streets and plazas below, we sometimes raised our eyes to watch also the calm, slow march of the great constellations across the sky above; glittering groups of stars that seemed to look down with cool and contemptuous eyes upon all this mad flurry of human excitement and human endeavor. Dozing a little now and then, we sat there until at last dawn sent its rosy light across the world. It was the last dawn, I knew, that Hilliard or I would look upon.

Now, it seemed, all the preparations in the giant air-city about us were completed. The crowds that had moved in its streets during the day and night before remained, but silent now with the thrill of approaching combat. Tense and silent the city remained, as the sun crept up toward the zenith through the morning hours of that fateful day. And, high in our tower-cell, Hilliard and I found ourselves gripped by the same tense feeling of anticipation. From our window as we watched the city, we made out the west, a dark spot rushing through the air toward Berlin, a spot that was

growing steadily larger in size, that was broadening out into a large dark disk; and then as it came swiftly closer we saw with astonishment that it was a city, a giant air-city almost as large as Berlin itself!

The Gathering of the Cities

We heard a stir of excitement in the streets below as that mighty air-city came closer to us; then saw it slowing down until at last it had come smoothly to rest out to the south of Berlin, hanging there in mid-air a half-dozen miles away. It was London! Even as I had recognized it, Hilliard had done so also. London! The great air-city that held all southern England for the European Federation, could be clearly recognized, not only by its size but by the somewhat different architectural design of its metal towers and plazas. We could make out clearly now the surface of the other city, its huge batteries of heat-guns, and its great towers surmounted by a central pinnacle. And now, as we scanned the horizon away to the north, we could see another dark disk, another mighty air-city, rushing swiftly toward us!

"They're gathering!" Hilliard's voice was agonized. "Gathering—all the air-cities of the European Federation! It's the beginning of the end."

"Gathering for their great attack," I said.

"God, if Connell and Macklin could have escaped!" Hilliard's cry burst from his tortured soul. "If our own air-cities had only the speed and the power to resist this attack!"

"Steady, Hilliard," I told him, my hand on his shoulder. "It's the end, I think—the end for our Federation as well as ourselves—but we must face it."

Now the air-city from the north was rushing closer, was hanging northward of Berlin, and we saw that it was Stockholm. And, even as it came to rest out there beside us, two other air-cities were rushing up from the south; looming larger swiftly and identifying themselves, when they too shot up to hang near our central city, as Geneva and Rome. And then from the west were coming others, Paris and Brussels and Amsterdam; while down from the cold east were speeding Moscow and Helsingfors and Leningrad. City

after city was rushing from all quarters of the compass, from every part of the European Federation, until they filled the sky. Through the hours of that afternoon we watched their numbers grow until they numbered over a hundred. They had come from every part of the earth, over which the European Federation held sway. From the bleak eastern steppes, from the jagged peaks and green valleys of the Pyrenees, from the great ice-locked fiords of the north and from the blue plains of southern Africa, they were rushing at colossal speed to gather here in a great circle about their capital city—Berlin! Great air-cities, each of which flashed through the air at the same tremendous speed, each of which bore upon it great batteries of those giant heat-guns that nothing else in the air could sustain, each of which held upon it a soaring electrostatic tower and thousands of other clustered pinnacles. As in Berlin, the crowding, seething millions of its peoples had been left on the ground quarters prepared for them. The gathering of the cities! At last, with the coming of sunset, all but the last few of the Federation's mighty air-cities had gathered around Berlin!

By then, gazing out from our window high in the electrostatic tower, Hilliard and I seemed to be looking across a single gigantic city that stretched in mid-air as far as the eye could reach, so closely were the scores of great hovering air-cities hanging together! It was as though we were looking forth across an endless plain of clustered towers of metal, from which rose here and there the higher pinnacle of a city's power-source; a titanic plain of towers and streets of metal, crowded with millions of the European Federation's soldiery. And, as the blood-red sunset flamed eastward upon all this huge assemblage, now waiting only for the last of their number, something seemed to snap in my brain, and all the stoicism which I had summoned to meet our fate and our nation's fate abruptly vanished.

"We *can't* stay here while doom rushes upon our nation!" I cried madly. "Since they start out tonight—since our time is up and we die ourselves tonight—we'll go to death fighting for our freedom!"

But, now, it was Hilliard who endeavored to calm me. "It's useless, Brant," he said. "A few hours more; then all will be rushing west while the Asiatic Federation is moving east upon our air-cities. And at any moment now, before that attack starts, they will be coming here for us."

"But they'll not take us to a death like that!" I exclaimed, a cold, long-repressed fury surging up within me. "If we're to die we'll do it, striking a blow at our enemies!"

Like a caged tiger I paced the little cell's interior, growing shadowy and dusky now; the sun had disappeared. From the corridor outside came the voices of the guards, and at any moment I expected the door to swing open and admit those who would take us to a last examination at which our silence would bring immediate death. Already, far out over the great mass of scores upon scores of giant air-cities that filled the air about us, a great, complicated pattern of brilliant lights was gleaming through the deepened twilight; and now, from south and east and west, the last of the great European Federation's air-cities were assembling about that tremendous gathered mass of cities. Then, as I turned from the metal door which I had been examining in blind and futile rage, my eyes fell upon our bunk-racks and the strong but slender strips of metal that held them out from the metal wall, against which they were set diagonally. And, as I looked at them an idea, a last flame of hope, burned into my brain, and I turned swiftly to Hilliard.

"Those strips of metal!" I exclaimed, pointing toward them. "Those bunk-supports—it's a chance to escape! A chance that means death, Hilliard, I think—but death is upon us now in any case—"

Swiftly, almost incoherently, I explained to him the idea that had suggested itself to me. I heard his breath catch as he comprehended its appalling nature. Then I saw his eyes gleam as he realized that, inasmuch as almost certain death awaited us, death in escaping could not deter us, for we were already doomed. So, we grasped one of the metal strips and tried with all our force to tear its lower end loose from the metal wall. That lower end, set directly in the wall, seemed integral with its metal; and, as we pulled upon the metal strip, gasping with our great effort, muscles tired, we still kept on. We had to work quietly lest some sound betray us to the guards without. It seemed that we could never tear it loose. Straightening from the violent exertion, with dizzy heads, muscles aching, we paused for a moment, then reached to grasp the strip again, braced ourselves against the wall and exerted all our force upon it. It held for a terrible moment, then seemed to give, to bend—and then, with a little grating sound, we had pulled the strip loose from the wall into which it had been set.

CHAPTER VIII

A Single Chance

Intently for a moment we listened without moving; but there came no sound of alarm from without, nothing but the occasional voices of the guards. And now we grasped another of the metal bunk-supports, and wrenched its metal strip loose from the wall with another tremendous effort. We had in our hands two metal strips, each of some three feet in length. These we now bent swiftly into two L's or right angles of equal sides, using all our combined strength on each to bend the strong metal. Then, swiftly loosening the long, strong leather belts that criss-crossed over our black air-jackets, we formed of them swiftly two leather ropes ten feet in length. Each of these we attached to one of our metal L's, making each fast to one of the jagged, broken ends of one of the bent strips. Then, panting from our swift efforts, we stood erect, and moved toward our little window.

Night lay over the world now, and from our window we saw the cities illumined by their lights stretching out to the horizon. On the landing plazas of the air-city beneath us rested the great European Federation battle-fleet. In the plaza directly beneath us, that which surrounded the base of our great electrostatic tower, there rested but a few score of cruisers, those of the commanders who were now at the headquarters inside the great tower's base. The plaza was practically deserted; for it was the evening meal. For a moment I stood there at the window, gazing out over that tremendous mass of giant air-cities. Then, summoning all my courage, I flung my right leg over the window's base, through its opening.

Sitting astride that opening, while Hilliard watched anxiously behind me, I placed the metal angle I carried upon the flat metal sill of the window, one end of its angle catching on the sill while the other end, to which my leather rope was fastened, pointed straight downward toward the great plaza a thousand feet beneath. Then, holding to that leather rope, I slid out of the

window's opening; and hung by my hands from the slender rope with only empty air between me and the plaza far below. Tensely I swung there in that moment, but the metal angle caught in the sill held my weight. And so, sliding down the leather rope fastened to it, I felt my feet strike in a moment against the sill of the window below. Another moment and I stood upon that sill, crouching within that window's opening.

The window in which I crouched opened into one of the great upper corridors of the electrostatic tower; but I knew that to venture back into the building, swarming now with guards, was to meet death, nor did I plan to do so. Giving my leather rope a twitch, I worked loose the angle resting on the sill above; and, when that dropped toward me, I placed it on the sill on which I stood, and the next moment was sliding down to the window below. And now above me Hilliard, using his own metal angle and leather rope in the same way, was following me, was sliding from window to window after me, down the smooth side of the mighty pinnacle to the street far below. Down—down—like two strange insects we crept downward from window to window. None in the streets below glimpsed the two tiny shapes crawling down the mighty tower's side; for the darkness had deepened now, and in the plaza directly beneath us there were none of the crowds that swirled elsewhere.

Our greatest danger, indeed, was that we would be seen by someone inside the tower as we swung down from window to window; and twice I was forced to hang for a few seconds from my leather rope above a window inside which I could hear voices. Yet still down and down we swung, praying that the regular line of windows in the static-tower's side, extended unbroken clear to its base; for otherwise we were lost. Down and down we went, moving more hastily now despite the awful hazards of our progress; so hastily that once Hilliard's hook or angle slipped halfway out from the sill upon which it hung, and all but precipitated him down to death before he could slide into the window beneath him.

But now we were within the last dozen levels of the plaza's surface, and were down with all the eagerness of renewed hope. For in the plaza there beneath us there lay still the unguarded cruisers, their officers and crews gathered in the great tower down which we were creeping. Another level—another—and down we swung through the dusk, in such a descent as surely man never had made before. The plaza was close beneath, the window of

our cell now far above. From far around that plaza, from Berlin and from all the air-cities about, we heard the great hum of final preparations being made. I knew they were ready now to sally forth upon their gigantic attack. But we were within the last few levels of the plaza, now, swinging down with mad haste from window to window toward its smooth surface. And then it was, when we were within a few yards of that surface, that I heard a dim cry from far above.

"Down now with all your strength, Hilliard!" I cried to my friend above me. "They've come to our cell after us—are giving the alarm!"

"The nearest cruiser there below!" he exclaimed thickly as he swung madly down after me. "We'll make it yet!"

But now the cry of our guards high above was being taken up and repeated by other voices in the great electrostatic tower. That cry was coming down through it from level to level, even as we swung from the last window to the level plaza. And, as we staggered across it, toward the open door of the nearest cruiser, there came a series of popping detonations from above and the next moment little flares of terrific heat were bursting all about us as the guards shot their heat cartridges down toward us! From their great height and through the dusk their aim was poor, and in a moment more we were at the cruiser's open door. But now the alarm was spreading over all the tower behind us, and at the same moment that we flung ourselves in through the cruiser's open door, slamming it behind us, we heard a wild clamor of voices from the power-tower's base!

The next instant, though, we were bursting up into the cruiser's bridge-room and for a moment of agony I fumbled at its controls, set differently from those of our American cruisers. Then the motor-stud had clicked beneath my fingers and, as the great electric motors beneath droned suddenly loud with the current rushing through them, I sent all their power into the cruiser's tube-propellers. Up it went rushing, up and away at terrific speed and at a steep slant, even as a mass of green-uniformed figures burst from the electrostatic tower into the plaza! Out and over that plaza at terrific speed we shot, out and upward at such awful mounting velocity that, before the great batteries of heat-guns around us could turn, before the alarm from the power-tower had time to spread, we were whirling up and through the

dusk over all the massed towers and gleaming lights of the great air-city Berlin!

Out and over—and now as we soared upward into the rarefied levels of the air like a shooting-star, our cruiser was driving outward over the cities, stabbing westward through the air, literally chasing the sun that had disappeared hours before. From Berlin behind us there rose a hundred cruisers, soaring in swift and deadly pursuit! But so swift had been our rush, so tardy had been the alarm of our escape, that before the great batteries of Berlin could blast us from the air we were beyond them; and before the other massed air-cities over which we were rushing could receive that alarm we had split the air westward above them, and had rushed out from over the last of their titanic floating masses and into the night!

"We're clear of the cities!" I yelled to Hilliard over the thunderous droning of our motors and the roar of winds about us. "If we can shake off these pursuing cruisers we'll win back across the Atlantic yet!"

"But their whole battle-fleet is rising now!" cried Hilliard, gazing back. "And now all their air-cities are beginning to move westward too—*all their hundred air-cities are moving west to the attack!*"

Across the Atlantic

Despite the wild peril of our rushing ship, I felt for a moment all the blood congealing around my heart as Hilliard yelled those words, and I looked backward for one last glimpse. For there, behind us, behind the hundred ships that were pursuing us, the whole two thousand cruisers of the European battle-fleet had risen and were coming westward also. They were not pursuing us so much as they were speeding westward according to their plan, moving after us in a great crescent formation! And, behind them, we could see now all the hundred gigantic air-cities of the European Federation, massed there in a colossal circular formation about their central city of Berlin; moving westward also behind the crescent of their fleet, they were flashing with terrible majesty through the air in their circular-massed formation; at a speed that mounted swiftly to two hundred miles an hour!

The great attack had begun!

Only a moment I gazed back upon that colossal spectacle never seen by man before, and knew that in that moment, far on the world's other side, the hundred great air-cities of the Asiatic Federation would be rushing eastward; the two great forces of hurtling air-cities were converging upon the American Federation. Then, as we shot forward with our own greater and terrific speed, the vast massed cities and the fleet before them had passed from sight behind us and only the hundred grim pursuing cruisers were visible in the night as we hurtled on!

On—on—and now I shouted to Hilliard to go beneath to the cruiser's motor room. He moved down toward them while I gripped the wheel tightly, standing there alone in the bridge-room of the long great cruiser that had but Hilliard and I inside it. And, while we hurtled on at our maximum terrible speed, as the cruisers behind drove steadily after us, we realized that we and our pursuers were outracing the sun around the earth! We saw by the growing light that we were high above the sea instead of land. The sea that we saw through breaks in the vapor-layer, gleamed to the west of us with sunset lines. We were over the Atlantic, and now, as hope of escape from our pursuers burned stronger within me, there came a sudden faltering in the steady drone of our great motors! I felt our cruiser lose speed in that moment, knew that faltering to be caused by the circuit-breakers tripping at the tremendous power we were using. But then after an awful moment of hesitation the motors were droning as loudly as ever as Hilliard, beneath, had thrown back the circuit-breakers in the connections that conveyed the static electricity from the atmospheric charge about us to our transformers. Only a moment had we faltered thus; but in that moment the hundred pursuers behind had come swiftly closer!

Onward still, like some phantom, we rushed, minute after minute of droning, racing flight, with the sunset ahead flaming brilliant now, as we overtook it. Steadily, the long gleaming ships behind us were creeping closer while the sun rose in the western sky. Though Hilliard was working like a madman in the motor-room beneath, tending the motors as a mother anxiously watches her child, he was but one and could not do the work of a dozen. And so, on after us, they came; drawing toward us so close that at last I knew that we could not win free of them in our frenzied flight. For, although we had rushed on for a time that seemed endless to us, and a few

hundred miles remained between us and the American coast, their leaders were so near now that another minute, I knew, would find them using their bow-guns upon us.

Even as the thought came to me, there was a thundering detonation behind us and then but a few feet to either side, a shining heat-shell flashed past us. Another detonation, and another followed, and I knew that not for long could we escape them thus; since with each moment the shooting was becoming more accurate. So, just as a dozen of their bow guns thundered again, I suddenly drove the cruiser downward in a flying headlong plunge down through the vapor-layer beneath us; and, as the pursuing ships plunged straight down after us, I sent our own cruiser instantly whirling upward and through that layer once more!

It was a maneuver that gained us a moment's advantage; since when the pursuing ships drove up through the vapor-masses again on my track, it took them an instant to locate and shape their course after me. In that instant we had moved a little from them; but now, remorselessly as ever, they came on after us, as pursued and pursuing ships drove like light toward the flaming western sky. On and on, on until again they were close behind, until again their guns were beginning to thunder, and then I repeated my former maneuver, my last resort. I dived headlong downward again through the vapor-layer, and upward again, as their ships drove after me. But, when I flashed up again through the vapor-masses this time, I suddenly slowed my ship, slowed it and then held it motionless there in mid-air, with cold of icy fear tight around my heart in that moment! For, this time, half of the hundred pursuing ships had not dived down after me but had flashed on ahead as I drove down; and so, now, when I flashed up they hung before me, while the remainder drove still toward me from behind! I was trapped at last between their masses of ships before and behind me!

Slowing my cruiser, holding it motionless there with mechanical fingers in that moment, I knew it to be the end. Our last moment had come. The two masses of ships were moving toward me, from ahead and behind, were moving toward our cruiser. Our only escape cut off, no twists or turns now could save us; they were converging upon us, were moving deliberately toward us from either side. Another moment, I knew, and hundreds of heat-guns would thunder from them, hundreds of heat-shells would send our ship downward in flaring, fusing destruction. Another moment—

A great cry sounded beside me, and I wheeled to find Hilliard pointing mutely upward toward a mass of long, gleaming shapes that were rushing headlong down upon us from high above, that were diving headlong down upon the European cruisers to east and west of us, raining a hail of heat-shells upon them! "American ships!" My cry was echoing Hilliard's. Great gleaming cruisers, outnumbering the hundred east and west of us, were driving down upon those hundred with all their heat-guns thundering! Then in the next moment, while our own cruiser hung motionless, helpless there in mid-air, American and European cruisers were whirling in a mad, swift battle about us, ships striking and falling like lightning on all sides of us. And before we could comprehend with our stunned minds what was taking place, the European cruisers had suddenly dropped from that battle, had massed together and were splitting the air eastward, rushing back eastward and disappearing toward the mighty approaching armada of great air-cities and cruisers of which they were the scouts.

Now from the mass of the American cruisers one shot toward our own, level and hung beside our own ship, and as its door was flung open we opened the door of our own. I stared at a tall figure, not crediting my eyes.

"Macklin!" I cried as I recognized him: "Macklin! You got clear, then?"

"Brant—Hilliard—!" he was himself exclaiming: "It is you two in that ship, then! We were sent on a last patrol out here, saw your ship attacked by other European ships, came to your rescue—"

"We escaped from Berlin," I told him: "But you, Macklin, we thought you and Connell dead, saw your cruiser struck by heat-shells and falling—"

"It was a last ruse," Macklin swiftly explained: "Those pursuing ships were overtaking us; so, when their batteries fired a storm of heat-shells after us, we fired one of our own heat-guns back toward them at the same time, and our shells meeting one or two of the oncoming ones made them burst and flare there behind us. Then, at the next moment, we sent our ship whirling down as though struck and destroyed."

"But Connell, then!" I cried: "Connell got back with his great secret. All the hundred European Federation air-cities are rushing across the Atlantic to the attack!"

We leaped across the gap to the gangway; the door of Macklin's cruiser closed behind us; and he gave the order that sent it, with the whole cruiser fleet; westward at swiftly mounting speed. Then he turned back to us.

"Connell got back with his secret, yes," he said: "And though the hundred European Federation air-cities are rushing westward and word has come that the massed hundred Asiatic Federation cities also are rushing eastward for the attack, they will find the great air-cities of the American Federation massed together and ready for them! In the ten days since our return, every effort of our cities has been exerted to make use of Connell's knowledge, and to equip with the new tube-propellers that will give them the same tremendous speed as our enemies. And now all our cities are massed together and waiting for the attack of our enemies."

CHAPTER IX

The Battle of the Air-Cities

Now the hundred cruisers of our force were cleaving the air westward at terrific speed, while Macklin and Hilliard and I stood together in the bridge-room of the foremost as it rushed on. Beneath us, the gray Atlantic showed here and there through openings in the vapor-masses, and ahead the sun still hung in the western sky. And within a few minutes more, we saw that the vapor-layer beneath was thinning, and that now we were flashing not over the sea but over land; over green hills and valleys that we could glimpse rushing past far beneath us. I gazed to north and south in search of New York and the other coastal air-cities that should have hung there, but nothing was in sight.

"All our American Federation air-cities," Macklin told me: "are massed together, hanging south of the Great Lakes. From Buenos Aires to Winnipeg, they've come."

"You think, then, that the European and Asiatic Federation air-cities are going to make a simultaneous attack from both sides?" I shouted to him above the roaring of our flight. He nodded emphatically.

"Undoubtedly. The Asiatic Federation cities are over the Pacific now, and are keeping in touch with the European ones by distance-phone to time their attack to coincide from east and west. They know our own cities have massed together, must know now that they've been equipped with the great new speed-tubes also; but they're coming on."

"Two to one," I said: "Two hundred air-cities attacking our one hundred. God, what a battle it will be!"

But now Hilliard had broken into our conversation, was pointing far ahead toward a dark, flat mass that stood out against the brilliant western sky, and

toward which we were moving. The terrific speed at which we had been racing on for hours was decreasing now. Far beneath the land was still rolling back at great speed, long green plains now; since already we had flashed west over the Alleghanies. Then, as the dark mass westward grew steadily with our approach to it, other ships were driving suddenly beside our own, watchful patrols that drove down upon our hundred cruisers and swiftly challenged them. Macklin answered those challenges by the distance-phone, but for the moment I paid small attention to him, gazing forward with heart beating rapidly at the great mass that hung high in mid-air before us. For, as we drove closer toward that mass, it was becoming visible to our eyes as our goal, the hundred giant air-cities of the American Federation!

The hundred mighty American Federation air-cities were clustered there miles above the green plains, in a great circular mass, with New York, most colossal of all of them, at the center! Cities that had long hung over North and South America from sea to sea, air-cities whose names were those of the long-vanished cities of the land, that once had dotted the surface of those continents. Boston and Los Angeles, Rio de Janeiro and Chicago, Mexico City and Quebec, Valparaiso and Miami—these and scores of others hung there in that great cluster. All the air-cities of the American Federation were gathered here about their air-capital of New York to withstand the tremendous attacks now closing in from east and west!

Massed there as they were, the hundred mighty air-cities seemed, even as the European ones had seemed to me, but one vast plain of metal towers and streets. As far as the eye could reach, there stretched away the tremendous forest of those soaring towers, with here and there rising from them the taller spires of each city's great electrostatic tower. And, everywhere among those towers, everywhere around the rim of each great circular air-city and at its center there loomed great batteries of giant heat-guns; while here and there, on the plazas of the cities rested the turret-like cubes of the recalled great air-forts, their own grim heat-guns protruding expectantly. And through streets and towers, between the batteries and around the air-forts and across the plazas of the assembled cities, there swarmed the millions of their peoples, wild with excitement now as the last dread hour approached. And, massed there above all the great floating

cities, hung grim and motionless the two thousand or more cruisers that still remained of the American Federation's eastern and western fleets.

It was toward these massed battle-cruisers, at a level somewhat higher than that of the air-cities, that our own hundred cruisers were rushing. Over those assembled giant cities we raced, the great mass of them below us almost hiding the ground beneath. As we shot above them, I saw now that they had been ranged in a tremendous circle, the great capital of New York hanging in the center. Across the great ring of the air-cities we rushed; were racing at last above New York, toward its own giant power-tower. Then we had reached it, and were sinking vertically downward, until our hundred cruisers came to rest upon the central plaza. Here, even as in Berlin, the central plaza was reserved always for the ships of the First Air Chief and his followers; so that, although immense crowds now beat through all the streets and plazas about it, there were none around our hundred ships. And in an instant Macklin and Hilliard and I were out of that which had brought us and hastening across the clear space toward the static-tower's base.

Preparations

On past its guards and through the ante-rooms we strode, and in another moment were in the office of the First Air Chief. There was unfamiliar apparatus among the great switchboards of its walls, I noted as we entered. The First Air Chief himself had risen from his great table-map as we entered and was coming toward us; beside him, another figure, whom I recognized instantly as Connell. Then both of them were grasping the hands of Hilliard and of myself.

"Brant!" Yarnall was exclaiming: "I got Macklin's report of your escape and his rescue of you—man, but I'm glad that you got free! And it was what you did there in Berlin, what you did to help Connell and Macklin escape, that has enabled us to use Connell's knowledge and fit our air-cities for the coming battle!"

"I did no more than the others," I told him: "But you know of the enemy's coming then? You know that already the European Federation cities are on

their way?"

He nodded. "They and the Asiatic Federation's cities from the westward, Brant," he said: "And we are awaiting them here—awaiting them with a chance at least, thanks to you four—to strike back at them when they come. And already they are near—by the map here you can see—"

And he turned toward the great table-map upon which was depicted the whole of the earth's surface, the red circles upon it denoting as before the position of the air-cities that hung above it. Now, however, all the circles of the American Federation cities were massed together south of the white outline of the Great Lakes, hanging motionless as the cities around us were hanging motionless. Away to the east on the map, though, just moving in from the Atlantic over the eastern coast, there was creeping across the map another mass of red circles, moving slowly toward our own, that represented the great gathered cities of the European Federation that were rushing westward toward us. And in from the Pacific was creeping a similar mass of a hundred little red circles that were, I knew, the Asiatic Federation's cities.

From east and west they were moving, there on the map, moving even as the cities they represented moved through the air, automatically showing their positions and progress. This was accomplished, I knew, by means of special batteries of cruiser-finders, tuned and trained to detect the great electrostatic-motors of air-cities, and recording instantly thus whenever those cities moved with their great electric fields. Their records were carried through complicated mechanical calculators which plotted the exact positions and movements of the cities; and these calculators, in turn, were connected to small special projectors set beneath the great ground-glass table-map, casting upward upon it the red circles of the air-cities. Thus those red circles moved upon the map, even as the great air-cities moved across the world.

This arrangement, indeed, was of no late date, and was used by both European and Asiatic Federations as well as by ourselves; but as I gazed now about the great circular room I saw that within it were some new arrangements also. These consisted of a series of six great glass screens which were arranged in box-like form about the great air-city's controls at the room's center. And, while the First Air Chief swiftly explained to us

their purpose and design, I saw that one sitting inside their box-form, with four on four sides, and one above and one below, could see in all those directions as though from the very top of the great power-tower. For they were in effect great electrical periscopes; four great similar screens had been set on four sides of the electrostatic tower's high tip, and another one above that tip, while the sixth had been set in the under-side of the great city's base. The views possible to those six screens were then transferred down to the six there about us; the light-vibrations that struck the screens above and beneath being transformed by television receivers into electrical vibrations and brought down to television reproducers behind our own six screens.

Sitting there at the city's controls, amid those six screens and looking into them, one could see as clearly as though from the power-tower's tip in all directions. It was quite necessary, too, that this should be the case; since the man who operated the great air-city, from its six direction-controls and its single speed-control there, must see clearly in all directions, now that the great air-cities could rush at such tremendous speed through the air. When I said as much to the First Air Chief, who had turned now and was gazing intently at the great table-map upon which the eastern and western masses of circles were slowly creeping toward our own, he nodded, and contemplated me for a moment with a curious expression.

"The man," he said finally, "who is to hold the controls of New York in the battle tomorrow, will be you, Brant!"

"The honor is great," I said: "I've operated the city's controls, though never did I expect to take it into battle. But Macklin and Hilliard here—I want them to stay here for the time being—I want the hundred cruisers on the plaza outside to stay there during the battle."

"You have a plan?" the First Air Chief asked, but I shook my head.

"No more than an idea," I said: "An idea that may help us if the battle goes against us, if their attack is too strong for us. Even then it is too insane, perhaps, to be of any use, but it might help us—"

Yarnall nodded assent, and then Macklin and Hilliard had joined the two dozen or more of black-garbed attendants and engineers who were busy at the great switchboards that lined the circular room's walls. They scrutinized

its dials to determine the rate of the vast currents rushing down from the power-tower's tip far above to the motors in the city's great base; added a fresh battery of transformers or threw in resistances to hold that current steady; and moved ceaselessly about the walls in their anxious watch. Now, Yarnall and Connell and I were marking our own places, the three metal seats there behind the big table-map, with the great screens of the electrical periscopes all about us. Yarnall would sit in the center, with eyes upon the red circles on the great map, tensely watching their progress, as admiral of our mighty fleet of colossal cities, ready to direct it and our cruisers to the battle. Connell would be at his right, before him the black mouthpiece and speaker of a single distance-phone. Behind that were the scores of switches and intricate controls, which connected that distance-phone to the operators of all our hundred air-cities.

The Battle Nears

As the third of the trio, I would sit at Yarnall's left, before me the six switch-levers which sent the colossal city of New York whirling through the air in any direction; while beside them was the gleaming knob which regulated the city's speed. The great batteries of New York were at my command also; all their mighty heat-guns around the city's edge and around our electrostatic tower and elsewhere were controlled by the distance-phone whose mouthpiece rose before me. The great batteries of all our other cities were controlled in the same way by their own operators, and were subject like New York to the commands of the First Air Chief beside me, who could maneuver our whole great armada of tremendous cities at will through the air. In the city of San Francisco, too, we knew, was the Second Air Chief, placed there to take command in case New York were destroyed or the First Air Chief disabled.

Thus, on the morrow were grouped we three, who were to sway such colossal forces in a battle as no men had seen before. Now, Yarnall was pointing to the table-map's surface, where the red massed circles of the European and Asiatic Federation armadas were indicated but a few hundred miles on either side of our own great mass of cities. Watching them there,

we sat in silence, save for the clicking of occasional switches by the engineers about us. From far away, far across New York and all the other air-cities gathered around it, there was coming the dull, dim throbbing of the life of millions that swarmed through those cities. And now Yarnall reached forward and touched the control of the great electrical periscopes whose screens boxed us in.

Instantly those dull-glass screens were alive with light, and it was as though we were gazing forth from the very tip of the power-tower out over our gathered mass of cities. North and east and south and west, from all the screens about us the views were alike, of a tremendous mass of clustered metal towers that encircled New York. Below us was the screen, above which our metal seats were suspended on supports. It seemed a trap-door through which we were gazing down toward the green plains far beneath; though in reality all the city's massive base lay between us and that view. So intensely realistic was the scene that lay about us that we all but forgot the great circular room in which we really were, and seemed suspended high in air above the great mass of our gathered air-cities.

"The enemy armadas," said Yarnall, his voice low, "will be in sight within fifteen minutes."

For upon the map the two masses of red circles were rushing on from east and west, and seemed now almost upon the mass of circles that was our own great fleet of cities. Looking out over those cities, through the periscopic screens about us, we could see the forts raising their great guns to firing range. I realized, as I saw it, that the battle now ready to start would mean annihilation to half the world. This was indeed Armageddon, when on earth itself was left no human being at peace; when every nation was rushing through the air toward this last conflict!

Now, however, Yarnall touched another control, and from the electrostatic tower's tip, high above us flashed great signals of brilliant lights that were taken up and repeated from all the power-towers of all the hundred cities that ringed us round. And, as those signals flashed, the great crowds that filled the streets of the air-cities were suddenly flowing out of those streets into the cities' towers; until within a few moments none were visible in all the streets and plazas, save those black-uniformed men who stood ready at the great heat-guns of our batteries. And those crowds went quietly, despite

their tense excitement, because they knew that they were being ordered inside for their greater protection. There was no refuge upon the earth's surface far beneath, for them; when the destructive powers of all the world were battling above it in the air.

Then the First Air Chief spoke a brief order and, as Connell beside him repeated it swiftly into the distance-phone (as he did with Yarnall's orders in all the combat that followed) the great fleet of cruisers hanging above us and visible in our top screen divided into two masses, of a thousand or more ships each, which swept swiftly to east and west. Beyond the great ring of air-cities they leaped, until they were far out; and each division then formed into a great curving line screening our ring of cities to east and west, facing the fleets rushing toward them from those directions. Then we were gazing again at the table-map before us, a deathly silence seeming to grip all the world. Upon that map we could see the European and Asiatic armadas were now within hardly more than a hundred miles of our own; and tensely we watched the east and western screens now, gazing out beyond our cities.

"They'll use their cruiser-fleets for their first attack," Yarnall was saying as we gazed tensely forth. "They'll try to wipe out our cruisers before they bring their cities on to attack ours."

I nodded. "It would give them a big advantage when the cities come to blows. But our cruisers beat them back once with the odds two to one, and now—"

I broke off sharply, and at the same moment heard a low breath from Yarnall and Connell simultaneously, felt seemingly a low tremor that seemed to run instantaneously across all our massed air-cities. For there, far to the westward, black against the sky, there had appeared a line of far-flung black dots that were growing very quickly in size, and that were massed together in a crescent formation whose horns were toward us. It was the advancing cruiser-fleet of the Asiatic Federation forces. Tensely we watched it as it came on; then we looked to the east to see a similar crescent of advancing dots, the European cruiser fleet. On they came, smoothly rushing toward our own lines of cruisers, hanging to the east and west of our cities; and then for the moment we forgot them as we made out, to east and west, behind them, advancing toward us, great black masses that even at that distance seemed to fill the air. They were the two massed mighty

armadas of the European and Asiatic Federation's air-cities, rushing to battle with our own!



CHAPTER X

The First Clash

For one moment we gazed toward them and toward the advancing cruisers that rushed before them, as though held by the grandeur of the spectacle. The tremendous mass of our air-cities hanging there, high above the earth the gleaming ranks of our own two thousand cruisers that poised to east and west, the advancing cruiser-fleets of twice their strength behind which there rushed gigantically on the great massed air-cities of our enemies—it was a spectacle breath-taking enough! But, then, the two enemy cruiser-fleets had come within a short distance of our own waiting cruisers; and, as they did so, both their fleets shot suddenly upward, as though in answer to a common order, to drive above our own cruisers and our air-cities!

Instantly Yarnall had uttered a swift order; and as Connell's quick voice sent that order flying out to our cruiser-masses, they too whirled upward and forward to meet the onrushing enemy fleets. Then, far out in mid-air to east and west of us the great cruiser-fleets had met, had smashed into each other with blind fury. Through our screens we saw cruisers enveloped by scores in blinding flares, in that first moment of combat, as they raged in two separate mighty battles.

But, east and west, our own cruisers were outnumbered two to one and, despite their fierce resistance, were being pressed steadily back toward our gathered cities. Closer and closer through the air toward us were reeling those struggling lines of cruisers, more and more American ships falling in white-hot destruction as the heat-guns of their opponents concentrated upon them. Still far beyond them, to east and west, the colossal masses of the European and Asiatic air-cities were now rushing on toward us! Yarnall sent another order flying out, and the next moment a mass of cruisers that had remained at the edges of our gathered cities, on either side, and had not been in the first battle, were leaping forward now, like suddenly released

hounds after prey. A score of them to the eastern combat and a score to the western one they flashed; and each, as it reached the lines of struggling ships, shot up over them and then flashed along their lines above them. And as they did, there shot downward from them over the battling cruisers intense jets of concentrated water-vapor, which puffed out instantly into great white cloud-masses that enveloped all the grappling thousands of ships!

Yarnall had brought into the battle, I saw, those great artificial cloud projectors that had saved us in our first great battle over the Atlantic! And, even as their white masses enveloped the struggling ships, he sent forth another order; and our own cruisers shot upward and downward out of those cloud-masses, at the very instant that they had been formed. Then, with all our own cruisers clear of the cloud-masses to east and west, and the European and Asiatic ships inside them driving for the moment in blind confusion, our cruisers poured a deadly hail of heat-shells down into the cloud-masses and the thousands of ships that swirled inside them! Only for a moment was it that they remained within the cloud-masses before fighting their way free; but in that moment they had been unable to fire a single effective shell at our own ships outside, and we had poured a veritable storm of deadly shells into the cloudy masses. So when, a moment later, the European and Asiatic ships broke from the vapor-clouds east and westward, our broadsides had taken toll of hundreds of their blinded ships, and hardly more than our own forces did they number now!

But, when they broke free from the cloud-masses and into open air again, they did not advance toward our awaiting ships; instead they shot back from either front toward their cities, in answer to some mysterious order. At once a similar order from Yarnall brought flashing back, to form again above our own cities, our own ships—still nearly two thousand strong and almost as many in number as the combined cruisers of our enemies. I heard dimly the great cheers that were rolling across all our cities, for this mighty blow which had beaten back our enemies' cruiser-fleets once more. Then those cheers died swiftly away; for, far away to east and west, the gigantic masses of the approaching air-cities loomed larger and larger, rushing through the air toward us!

Eastward the European Federation was approaching, in a gigantic circle that seemed to fill the whole eastern sky. To us three sitting there amid our

periscopic screens, it was as though all the eastern screen was filled with that great oncoming mass of cities: Paris, London, Moscow, Cairo, Rome—all the tremendous air-cities of the great European Federation—and their capital of Berlin still at the center.

And westward, upon our western screen, loomed equally gigantic the similar circle of air-cities of the Asiatic Federation. Peking, the third in size of the world's air-capitals, at the center of its cluster of cities rushed at the same smooth speed toward us, with Shanghai and Tokio and Bombay and Rangoon foremost in its circle. From east and west thus the two stupendous circles of cities rushed toward our own, grimly waiting; and, as we watched them, there flashed over me in that tense moment a strange wonder as to the feelings with which one of a hundred years before would have watched this battle. Then I laid aside that passing wonder as the two great circles drew nearer to the range of our great heat-gun batteries. And this, surely, was a spectacle that none had ever seen or dreamed before—this spectacle of the world's mighty cities converging swiftly upon each other in battle to the death!

And as strange, too, must have been the sight of us three, sitting there amid our box-like periscopic screens, the heart of the mighty tower's base; yet seeing and directing from there all the great mass of the hundred mighty cities around us.

"They're not going to join forces!" I exclaimed: "They're going to strike us at the same moment from east and west!"

Yarnall nodded, his eyes intent upon the screens: "Either that," he said, "or —"

Armageddon!

He broke off suddenly; for at that moment there came to us a giant salvo of thunder from both east and west, a terrific shock of sound that rolled deeply through the air toward us from the two distant armadas advancing! An instant later there was a whistling sound over our own cities, and we knew that a great storm of heat-shells was plunging toward them. A few struck

with wild bursting flares across our massed cities, and with their suddenly-released gigantic heat there appeared great craters of fusing, melting metal. The greater part of the shells, though, fell short, whirled down to earth to flash on the ground far below. The range was not yet closed, I knew; and so far our own great cities remained silent as death. Yarnall was watching now with hands clenched tightly, as the two circles from east and west came on. A moment later there rolled from them another great salvo, and another mass of shells rushed toward us; but, though most of these also fell short, a greater number than before flared and fused in melting, searing death upon our own massed cities!

I gazed anxiously at Yarnall, a strange dread taking possession of me for an instant as the tremendous armadas came on. Would he never give the order to fire? He was sitting still as a carved statue, his eyes upon the screens and his lips compressed; and in that moment there came to me a dim sense of all the terrific responsibility that weighed upon him, the leader of a third of earth's cities and peoples battling against the remainder of mankind. Silent he sat and still, while the great approaching armadas rushed nearer, their cities coming more and more distinctly into view in our far-seeing screens. In another moment, I knew, another thunderous broadside would be belching toward us. But, just as I looked for it to come, Yarnall spoke a single word and, as Connell's swift voice sent that word flying out to all our cities, as my own flashed it to all the batteries of our center-most city of New York, there was a hush of a split-second. And then, out from all our own hundred giant cities, there broke such a titanic thundering detonation as seemed to shake violently all the air about us! And, an instant later, we saw the heat-shells from our batteries falling in thousands upon the two advancing armadas, upon their rushing gigantic cities, and flaring into white-hot craters of fusing metal.

Before that terrific blow the two advancing masses seemed to stagger a little in mid-air, to hesitate for an instant; but then they advanced steadily onward and their own great batteries thundered an answer to our salvo. Then, as the two giant circles drew nearer to our own on either side, the whole world seemed swallowed again in one ceaseless thundering of sound. The giant batteries of all our great air-cities burst forth again, to send a storm of heat-shells rushing east and west upon our enemies! The air between us and the armadas nearing on either side must have been filled

with shells in that moment; for we saw tremendous flares there in mid-air as shells of ours met some of theirs and burst. But few of them did that, most of them whirling through the air to burst and flare in all their awful destroying heat and brilliance upon the air-cities!

In that instant, as the batteries of all three gigantic armadas thundered, it seemed that numberless fountains of brilliant light and terrific heat were springing into being amid the metal towers of all the three masses of great air-cities! Great fountains of heat unthinkable, beneath which all the metal about them fused and melted instantly, and about which all life was scorched into annihilation! Yet this had taken but the first moment of the battle to accomplish; and now, as the two great circles of the European and Asiatic Federations swept nearer, I saw that two of the foremost of the great European cities, Paris and Lisbon, were staggering and reeling as they rushed forward with the rest! They had met an awful storm of striking heat-shells from our own cities; and though their great motors were deep in their mighty bases, though the alloy of their electrostatic towers could not be affected by heat-shells, yet the deadly hail of shells that had fallen upon them had penetrated apparently almost down to their great motors beneath!

I whirled to Yarnall, "Paris and Lisbon!" I cried: "They're falling behind the rest a little—they're hit hard!"

He nodded, eyes burning now: "All east batteries concentrate on Paris and Lisbon!" he ordered; and, as Connell's voice sent the order flying out to our eastern massed cities, their batteries were thundering with even greater fury.

But now all their shells were aimed toward the two crippled cities of Paris and Lisbon, and a hell of bursting shells were flaring across those two ill-fated giants of the air. It was as though a living wave, of brilliant heat and light from the bursting shells, were dancing like lightning across the two cities; and in that instant it must have burned from them all the life in their towers, even as they were striving to do with our own. The towers of each city became in that moment almost a single great surface of fusing, white-hot metal; its awfulness added to by the exploding of the stores of heat-shells and magazines beneath the terrific flares. Out beyond them and around them there swept other cities of the European armada to protect them. There came Madrid and London and Moscow, all their guns thundering answer; but now our own gunners were not to be denied of their

prey. As our batteries sent repeated storms of shells upon the two doomed cities we saw Paris staggering, slipping to a lower level, hanging a moment there, and then whirling sidewise downward, down to the green earth far below! In a moment it had crashed, with a terrific rending and cracking of metal!

I heard a dull roar of cheers rolling across our city above and between the thunder of the guns, cheers that were redoubled a moment later as Lisbon too whirled downward to annihilation, a mere mass of fusing metal. But, even as the cheers sounded they were stilled, and Yarnall was uttering further orders to our batteries.

From the Asiatic Federation's circle, a great smothering fire of shells had been falling in those moments upon the westernmost of our own cities, upon Omaha. Its guns were still thundering in savage answer; but, battery by battery, they were going silent as there fell upon them the concentrated fire of the whole advancing Asiatic Federation cities, the great guns of Peking thundering at the center of the rest. Then Omaha, too, had slipped and staggered and was whirling down to earth in fusing destruction, its motors reached by the bursting shells at last!

Avoiding the Ambush

Now it seemed as though in all the world was nothing but thundering guns and bursting shells; for now as they came nearer toward each other the three great circles of cities were exchanging a veritable tempest of heat-shells upon each other. Watching that hell of battle through our screens, we three sat tensely there. Yarnall's eyes were intent upon the advancing armadas; Connell, gripping his distance-phone, was barking orders to the great cities that were thundering about us. The two giant circles of the enemies' cities were very near our own now, rushing toward us at their tremendous speed; and, as they thus neared us, it seemed that nothing could surpass the tremendous roaring broadsides that were hurled from city to city. I saw Amsterdam and Madrid staggering a little behind their fellows, reeling beneath our awful fire, saw Hong-Kong in the Asiatic forces, its great towers all but levelled by the flaring heat-shells, plunge suddenly

downward as more of those shells reached its motors! But in our own mass now New Orleans was plunging likewise beneath the fire of the advancing fleets; and St. Louis was swaying as though badly hit!

But at that moment there came an abrupt exclamation from Yarnall; and then we saw that the two advancing circles of the enemy cities, rushing toward our own, were changing their form, were changing swiftly into two great crescents of which the horns of each were toward us. Those two giant crescents were moving to join each other, to form one great circle; and, if they did so, our own mass of cities would be completely surrounded by the overwhelming numbers of our enemies, the easy target for all their mighty batteries. We would inevitably be annihilated by the enclosing circle of the enemy. But, even before I had understood that maneuver, Yarnall's swift order was flashing out to all our gathered cities.

"North at full speed for all cities! Unchanged formation!"

The next instant our whole great mass of cities was moving, was moving with swiftly mounting speed northward! For, as his order sounded I had jerked open the speed-control before me, had flung back one of the direction-levers. And, as I did so, there had come a great droning of motors from beneath, resounding even above the madly-thundering guns, flinging all the mighty city of New York northward. Also there came the great hissing of the numberless new tube-propellers that were jerking it swiftly forward! And, as New York leaped northward at my touch, all its great batteries still detonating, so were all the great air-cities that ringed us leaping northward, and all their guns were still thundering toward the advancing armadas!

But now the enemy had seen our swift leap northward, and, as their commanders guessed our purpose, they sent their own two crescents whirling toward us with even greater speed to enclose us before we could escape! For the next moment it was a race between the three great city-masses, a race in which our own sought to evade the two that closed upon it from either side. And, as our cities and theirs raced through the air at tremendous speed, every gun still firing, it seemed that we must lose! For, just north of us, the two northward horns of the closing crescents had almost met, were almost joined before us! That northward horn of the Asiatic crescent held Shanghai and Colombo and Singapore and others;

while the horn that projected from the European mass had foremost in it Moscow and Brussels and Algiers. And as we shot northward in that wild moment, to escape before those two horns could join, Yarnall sent flying forth a swift order for all batteries in all our cities to concentrate their fire now upon the foremost cities in the two closing horns to the north!

At once our own guns were thundering with redoubled fury; for, unless we could destroy in the next few minutes the foremost of those cities north of us, they would have closed upon us and brought our irretrievable doom. So, disregarding for the time all the other air-cities of the two closing crescents about us, all fire was concentrated upon the foremost cities of the two horns closing northward. A storm of heat-shells rushed thick through the air toward them; but at the same time the masses of European and Asiatic cities east and west of us were pouring down upon us the broadsides of their own giant batteries! And beneath that terrific fire, cities among our mass were falling swift in fusing destruction. St. Louis and Miami and Seattle were whirling to death as we raced onward; all the people in them who had been left alive by the shells were meeting annihilation in the great crash far below!

But, though we were being decimated by the fire of the closing crescents on either side, our own terrific concentrated fire was having effect upon the closing horns of cities north of us, and in the moments while we rushed toward them, Singapore, Colombo and Brussels had been sent down in white-hot destruction by our awful fire. The remaining cities in those two projecting horns were still rushing toward each other with their utmost speed to close the gap between them before our great circle could speed through it. With Moscow and Shanghai at their eastern and western tips, the two horns swiftly closed toward each other; while as swiftly and with every motor droning its loudest, with every heat-gun thundering northward, we shot onward. For a moment the whole great race was in doubt; for a moment it seemed to us that our great mass of cities could flash through that gap before it closed.

But as we watched in tense, terrible hope, even as our mighty cities raced northward, the cities of those closing horns seemed to make a last supreme effort, a last great burst of speed. They shot forward, their leaders Shanghai and Moscow almost racing into each other; and then, with all their tube-propellers reversed, they were suddenly halting a barrier of mighty air-cities

all around us! But nothing now could halt our tremendous mass, so awful was our speed and so close were we to the enemy's line. I saw those cities looming suddenly gigantic before our own mass as we raced on; heard hoarse exclamations from Yarnall and Connell beside me; and then with a terrific shock, that seemed the shock of meeting worlds, our vast northward-flying mass of air-cities had crashed headlong into the great line of cities before us!

CHAPTER XI

City Against City

In the moment that followed, after our flashing circle of cities had thundered headlong into the line before us, all other sounds, all the thunder of countless guns and the drone of motors and the hiss of tube-propellers and cries of voices, were drowned in one tremendous splintering crash of metal upon metal. The giant mass of cities about us seemed to reel drunkenly in mid-air in that moment. New York at their center staggered from the awful shock transmitted to it. Then upon the screen north of me, I saw titanic metal masses that had been cities falling downward. Moscow with Sydney and Algiers, and Boston with Detroit, these whirled downward in that moment—no longer recognizable as air-cities and seeming to the eye but great twisted bulks of rended metal.

But not even that giant collision had been able to halt the tremendous momentum of our northward-thundering mass of cities; for as those cities before us crashed downward, the whole great circle of our mass, New York still at its center, was thundering on through the gap that that crash had made in the line before us! We were sweeping northward and out beyond that line that our great crash had shattered. We had won free only by means of that awful crash. Instantly, Yarnall had cried another order, and our great mass of air-cities was swiftly shifting its formation into a long line; and at its head rushed our own city of New York. And then, while the great circle of the enemy's combined armada remained there for the moment still behind us, as though stunned by our colossal crash and escape through it, the First Air Chief sent another order flashing forth. At once our line turned like a wheeling snake, high in the air, and was rushing back upon the circle of our enemies! We were rushing back and along the line of European Federation cities that made up that circle's eastern half for the moment. And, as our

long line of mighty cities whirled past them, all our batteries were thundering upon them roaring death.



Now our line turned like a wheeling snake, high in the air, and was rushing back upon the circle of our enemies. And as our long line of mighty cities whirled past them all our batteries were thundering.

The advantage now was all with us; since in their great circular formation more than half their great mass of cities could not reach us with their guns. And, so during that moment, the odds were more than even as our long line

swept on, with all our batteries pouring their broadsides forth! Steady at the controls, I held New York at that colossal line's head, holding it at an even distance from the great circle of the enemy cities. I saw now that our terrific fire, as we rushed past those cities, was swiftly in that moment taking effect upon them. Even as I gazed, Amsterdam, Vienna, Cairo and Madrid were falling beneath the awful concentrated fire of our rushing cities; and in our own line Buenos Aires staggered, swayed and fell as the enemy barked savagely back toward us.

"On!" Yarnall was crying beside me: "It's our chance to strike hard at them—before they can bring the rest of their cities into action!"

"But they're doing it now!" I shouted back to him, above the thunder of guns, the drone of motors and hissing tube-propellers. "They're stringing their circle out into a line also!"

For it was plainly visible, upon the great screen beside me, that the commanders of the enemy were striving to form their great circle into a line that could meet our own more effectively in mid-air. We saw their cities rushing inward and changing formation there beside us; but we knew that our change had come and so hung upon the flank of the great mass of cities, our line rushing along it with all our guns turned toward it. And now, though Quebec was falling in our line, our guns had sent down Copenhagen, Yokohama and Calcutta in the mass beside us; since by now we had raced past the mass of the European air-cities and our guns were thundering against the Asiatic cities. Their guns roared sullenly in answer to us as we flashed past them; but, for the moment, in their disorganization, in their attempts to reform swiftly from their circle into a line like our own, their fire was seriously hampered by their own movements. In that moment we were pouring a smothering hail of shells upon them, and city upon city was whirling downward in wild destruction.

Then suddenly, with a supremely swift effort, their circle had lengthened, straightened, their confusion of the moment had vanished; and their cities had formed almost instantly into a long line, like our own, but longer than our own. We found ourselves in that moment with our own line parallel in mid-air to theirs, a mixture of European and Asiatic cities directly opposite us; and then, as we raced on, they were racing on with us, their own batteries thundering with newly-released fury, as they sought to blast us

from the air beside them! I heard the sharp order of the First Air Chief beside me, and held New York steady at the head of our line. The two great city-fleets were racing through the air in a great running fight, with every gun thundering!

Directly opposite New York there raced, at the head of their line, the mighty air-city of Peking, third of the three great air-capitals. The two giants were evenly matched; and now at the head of our respective lines we engaged in a tremendous duel in mid-air, a duel so intense that, almost, I forgot the fate of the rushing armadas behind us. In all New York about us mighty fountains of brilliant light and awful heat, were flaring with each salvo of heat-shells from the Asiatic capital. But, at the same time, our own gunners were working their batteries like madmen; and we could see similar giant craters of fusing metal springing into being over all the vast mass of clustered towers that was Peking. Far behind in the enemy's line raced the third great air-capital of Berlin; and we guessed that it was now from Berlin that the movements of all our foes were being directed. But so awful was the battle that we were undergoing with the Asiatic capital opposite us that for the moment we forgot almost all else.

Behind us, I was dimly aware of all our great line of air-cities grappling with the line rushing opposite; Chicago, a little farther back in our line from ourselves, was carrying on as terrific a duel with London. Constantinople in the enemy's line was whirling downward beneath the batteries of Denver and Valparaiso. Montreal, in our own, was falling in fusing death as it became the target for all the giant batteries of the colossal city of Berlin. City after city in the two racing, struggling lines was falling to annihilation as the awful battle raged on. High above the colossal lines of racing cities, our own great fleet of cruisers and the enemy's were whirling in a wild fury, insane as our own giant battle of cities. Surely Armageddon had come upon the earth at last!

Although the European and Asiatic cities still outnumbered us, we had cut down their great margin of superiority in that attack which our line had made upon their confused circle. Now, with equal fury, they were striking from line to line.

Straight ahead of our two racing lines, there loomed now a great bank of drifting vapors, great cloud-masses drifting south from the lakes to the

north. Neither of our two battling lines desired to enter those vapor-masses, and so as one, when we neared them, both lines shot downward.

Surrounded Again

Down—down—with our batteries thundering still across the gulf toward Peking, whose own guns answered with as great a fury, though in their city as in ours, battery after battery was being silenced! Down—down—until the green earth beneath, lit by the descending afternoon sun, seemed just beneath us, rushing up toward us with awful speed as we shot down to it! Yet in that dread moment neither line of struggling cities straightened upward, each fearful of the other's gaining an advantage. In an instant more, it seemed, New York must crash together headlong into the earth.

Downward we shot, and I saw the green plain looming awful beneath us. At the last moment I jerked back another of the direction-levers before me; and as, in answer to the controls, New York tipped sharply upward once more. I saw Peking opposite rushing up at the same instant, saving itself as we had done. Behind us the cities that directly followed in our two lines were curving up as swiftly, all their guns thundering still as furiously. But, farther back in the two lines, there were cities that had swooped too low to recover, had dipped lower and in the next moment had crashed and been annihilated upon the green earth, as they collided with it at their terrific speed!

But while our two lines were whirling upward at as steep a slant as they had descended, the battle seemed to deepen in furious intensity. New York and Peking were stabbing still at each other with all their forces, each colossal city seeming too mighty to be struck down, though each was flaming under a terrible fire of shells. Behind us, after a running duel that had achieved almost the magnitude and fully the intensity of our own, Chicago had given London the finishing stroke; and that great city was wavering, staggering, then slipping and falling in white-hot annihilation toward the earth! And, as all along both lines, other cities staggered and fell, I saw that above us the two whirling cruiser-fleets had almost entirely vanished. They had almost entirely annihilated each other by the insane fury of their attacks!

I felt my brain spinning, felt all things about me resolving into a wild whirl of thunderous sound and flaring light. I heard, as though from a great distance, the orders of Yarnall beside me, and the frenzied voice of Connell sending those orders flashing out from the distance-phone; heard the thunder of guns and sound of motors and propellers and wild noises that were coming from all the city about us. Then, before our two onrushing lines, there loomed another great mass of drifting vapors; and again our two lines dipped downward to avoid those masses. But, as we shot downward, the line opposite us with Peking at its head shot as quickly upward again, in answer to some command; it raced on *through* the vapor-masses instead of beneath them!

In the next moment we had shot our own line upward again to race side by side with them still; but we were too late, for that moment had given them, at their tremendous speed, the advantage for which they sought. For in that moment, rushing on through the vapor-masses instead of beneath them, they had gained a little; so that when we shot upward again to their level they were ahead of us as well as beside us! And then their line ahead of us was swiftly curving back and around our other side! As we slowed instantly, to avoid a collision that would annihilate us and all our mass, they swept the end of their longer line around the rear of our own, and thus in the next instant were forming a complete circle around our cities. They had at last accomplished their great objective, had managed to surround our mass of mighty cities, outnumbering us still.

As their circle closed lightning-like about us, we three sat in that moment as though stunned; and then, from all the air-cities that encircled us, a terrific thunderous fire was pouring upon us! Encircled, we were a perfect target now for all the European and Asiatic gunners around us, pouring all their mighty broadsides upon us. And now Yarnall had leaped to his feet, the tense agony in his eyes reflected in those of Connell and myself in that terrible moment.

"They've got us!" he was crying hoarsely: "They've got us inside their circle at last—they're hammering us to doom!"

"Can't we break out?" I cried: "Break through this circle about us?"

He shook his head, his eyes burning: "No—their circle is complete around us now and we'd only crash our own cities to earth—but we'll try above and

below!"

With the words, he gave a brief order and, as Connell's voice flashed that order to all our confused mass of cities, they leaped upward in sudden concerted motion, all their motors' energy turned suddenly into their vertical lifting power. But, as they shot upward thus, to win free of the circle about us, that circle lifted at the same speed as our own mass, hovering still around us and beating us still with all the relentless fire of their massed batteries. And, when we shot suddenly downward in an attempt to escape from below they sank downward at as quick a speed, were encircling us still. And now, beneath that awful hammering fire of all the massed cities that enclosed us, our own were beginning to stagger; to sway and reel!

The titanic circle of enemy air-cities about us and our own great throng of cities, each a giant circular mass of belching flame, floating there miles above the earth; the thunder of each city's giant batteries, and the terrific brilliance of the storms of heat-shells that struck from city to city; the great glowing craters of metal that each striking shell made in a city, all these things seemed merged in the six periscopic screens that enclosed us like some chaotic and meaningless panorama!

I was aware of Yarnall's agonized expression, as we strove with every power that was ours to save our great air-cities from destruction. For now in our cluster, city after city was falling beneath that deadly fire of fusing shells. Los Angeles, Winnipeg, Panama, and Nashville whirled down one after the other. And, though our batteries were still thundering their roaring answers, our surrounded cities were still striking savagely out, with the colossal batteries of New York still thundering loudest, I saw how swiftly we were being annihilated! For, raging there in fearful battle high in the dusk between earth and stars, there was left now hardly more than sixty of our hundred air-cities; while in the circle about us there still hung, despite the giant blows we had struck them, a hundred or more of the European and Asiatic cities! And with all their guns thundering into us, the odds were swiftly changing and becoming more in their favor!

Finally I stood up, as though jerked to my feet by some strange force greater than myself, and wheeled toward the First Air Chief.

"It's the end now, Yarnall!" I cried to him above that thunderous roar of battle that seemed splitting all the night about us: "The end for all our cities

within an hour if this keeps on!"

"The end!" he said, his own face grim: "But there's no escape—we can only meet it fighting!"

My eyes held his fixedly in that tense moment. "The hundred cruisers in the plaza outside!" I said: "The cruisers I had you keep waiting for me—that last crazy plan of mine is our one chance now!"

"Your plan?" he cried, a flicker of hope rising into his eyes. But when I explained that plan in a few swift words his eyes widened with sudden stunned astonishment, and he cried out: "The thing's insane, Brant! You'll never make it!"

"But it's our last chance!" I shouted to him as the thunderous drumming of doom all about us deepened, and two more of our cities crashed earthward. "It's the one last chance to save our cities!"

He paused there silent a second, then reached, wrung my hand tightly. "Then go, Brant!" he said simply: "Take the hundred cruisers—and God grant that you are able to do the thing!"

I shouted to the black-jacketed attendants who were working like madmen around the great room's instrument-panels, cutting out motors that the heat reached, switching in spare motors and tube-propellers, keeping the mighty sustaining power of New York steady. Two of them leaped swiftly at my call, to the side of Yarnall to take the places of Connell and myself. And then Macklin and Hilliard who had been working with them, were running toward us also, and we four were running across the room and through the ante-rooms until we issued out from the electrostatic tower's base into the great plaza.

Standing there in that plaza with the darkness gathering about us, there stretched from horizon to horizon a boundless mass of gigantic light-gemmed cities, our own and the titanic ring that encircled us. The myriad lights of those cities, though, by which their gun-crews worked the great batteries, were feeble in comparison to the tremendous and blinding flares of brilliance in all directions that were fountaining up in giant gouts of dazzling light. Across all those cities floating, there leaped and flew the heat-shell flares, and the thunder of the guns was deafening, titanic, like the thunder of a stream of falling worlds! Beneath that thunder there came to us

thin, high cries, the wild cries of crowds in the towers of cities and in their streets and plazas. And high, high above all these, far, far overhead, began to gleam the pale mocking eyes of the distant, watching stars.

All about us, in that moment that we burst out onto the plaza, it seemed that heat-shells were striking and flaring. But the static-tower itself was of a composition that the shells could not harm, the rare refractory alloy that in air-cities is used only for the vital power-towers. And, though shells had struck here and there at the plaza's edge and on its surface, though there were on it and around it still-glowing craters of fusing metal, few of the hundred close-massed cruisers that waited with their crews upon it had suffered serious injury in the awful course of the battle so far. And now Macklin and Connell and Hilliard and I were racing across the plaza toward those cruisers, into the foremost of them and up to its bridge-room. Then Macklin jumped to the controls, as I called an order into the distance-phone over the titanic drumming of guns. In the next moment our hundred cruisers were driving up like mad things above the titanic battle raging there above.

Up—up—through a wild inferno of rushing shells, up over all the struggling, thundering, reeling cities we sped, on the wild venture that was our last wild chance. As we drove upward, I now saw others of our central mass of air-cities falling. Atlanta and Cleveland and Mexico City were whirling downward, giant masses of lights in which glowed countless great fusing craters of metal, gyrating insanely down through the darkness to crash in awful destruction on the surface of the affrighted earth far below! Hardly more than a half-hundred, indeed, remained of all our air-cities now; and the odds against them swiftly lengthened, as they were hammered still upon an anvil of fire and death by the circle that hemmed them in. They were staggering ever swifter, were reeling and swaying so that within a few minutes, even as I had said, the remorseless fire from all about them would send them to earth also, and wipe the last of the cities and peoples of the American Federation from above the earth!

But, as I saw that, our own cruisers were whirling on above that giant central mass, toward the great ring of enemy cities about them. Macklin at the controls, with teeth set, sent our ship and those behind it driving low above the awful combat with the storms of rushing shells from both sides thick about us. Ship after ship behind us was flaring and fusing and falling in white-glowing meteoric destruction, unnoticed and unheeded by any in

the titanic thundering battle beneath! On—on—we sped, rocketing through the night, seemingly the only cruisers now in the air, since the two great fleets had all but annihilated themselves. Yet as we shot on, it seemed almost that no cruisers could exist in the air over that great battle; since in dozens, in scores, our own ships were falling, stricken by the tempests of shells through which we were rushing!

But now we were reaching our goal, the giant Berlin that hung there in the enemy's circle with all its mighty batteries thundering again our doomed cities. Down toward it our cruisers swiftly rushed, unseen by any in the wild confusion that swept that city below us; down until we saw plainly the terrific spectacle of thundering batteries and wildly-rushing men. Here and there were heat-shells bursting and flaring in dazzling death, as the guns of our own cities roared savage answer. Down toward it moved our ships, now hardly more than a score in number, until there loomed just beneath us, that mighty central static-tower in which we had been so recently imprisoned! As we shot down toward it I beheld a glass ball above its tip, recognized that as similar in purpose to the periscopic screens on our own tower's tip, and then we had shot down past it, until our score of cruisers hovered beside the great tower's side, at the fifth level.

Hovering there in that moment beside the tower, with all the wild confusion raging beneath, and the plaza below still empty, we were still unseen by any beneath, by any in the great batteries that were thundering all around that plaza. Poised there in the darkness, we could see that the windows beside us were bright-lighted; that guards were swarming in the static-tower's upper levels, rushing to and fro. Then as our cruiser's door swung open, just level with a window beside us, Macklin and Connell and Hilliard and I were springing forth from that door and across the narrow gap, through that window, our heat-pistols ready in our hands! And at the same moment there burst after us our crew, and from all the windows around that level, from all the cruisers hovering beside those windows, a stream of black-uniformed Americans with heat-pistols in hands were pouring into the tower's fifth level!

Instantly the guards in that level were snapping their own weapons up toward us; but before they could fire a score of cartridges from our pistols had flicked and flared among them. As they sank lifeless in scorched, burned heaps of flesh we were racing through the other rooms and corridors

of that level, killing the guards in it with our heat-pistols, the surprise of our attack taking them unawares. So awful was the drumming of the titanic battle all around and outside, that no alarm of our presence penetrated to the levels above and below us, and, now with the last of the guards wiped from that fifth level, I turned toward my three companions.

"Connell and Hilliard! Take half our men and find your places here in the tower, keeping anyone in its upper levels from getting farther down than this! Macklin, watch with our cruisers outside—at this low height the batteries around the plaza can't reach the ships, can't pivot toward you—and be ready to keep anyone from getting into the tower from outside!"

As they whirled to obey my orders with the other half of our men, some hundred and fifty in number, I was running toward the cage-lifts. With swift blows we destroyed the controls that guided them from level to level of the tower, and then we rushed toward the narrow stairs that led also downward. Another moment and we were rushing down those stairs, while as we did so there came a scuffle of battle above us, and we knew that the alarm had penetrated to the upper floors of the tower and that the guards there were pouring down to battle with Connell and Hilliard and their men. We leaped on downward, though, down until we had burst down into the fourth level. There our surprise was as complete, and before the guards there were aware of our presence, almost, we had sent our heat-cartridges flaring among them, had swept them from existence and were leaping down to the third level. And in that it went the same; and in the second below it, and then, with hearts pounding, we were rushing down into the first level!

As we poured down into its ante-rooms, its guards rushed toward us and their own heat-pistols came up; but they too were falling in scorched heaps a moment later, and we were dashing through the ante-rooms toward the great circular inmost chamber that held the inmost controls of this great air-city of Berlin! Through those ante-rooms we burst, the surprised guards falling lifeless and burnt before us, and then into the inmost circular room! All around its panelled walls moved green-uniformed attendants who whirled with surprise from their switches and dials at our entrance; while at the room's center there was what seemed at first to be a great dull-glass globe! We knew that that globe enclosed within itself the great table-map and controls of Berlin; and now there were bursting out, through an opening in that globe, the three green-uniformed men who had been within it—the

First Air Chief of the European Federation and the two officers who with him had been controlling the movements of all the mighty combined European and Asiatic cities!

The Captured City

As the leader saw me, his swarthy face lit for a moment with a flash of recognition, of astonishment; and then he and his fellows were leaping toward us, their hands flashing down toward the heat-pistols at their belts as the attendants around the room jerked forth their weapons also. But as they did so, our own heat-pistols flashed up and for the next instant the great room seemed full of flares of blinding light as the cartridges burst among them, sending them staggering and swaying and falling in seared heaps! I shouted to my men swift orders that sent a score of them to the great switch-panels to take the places of the attendants there; while the remainder rushed toward the great doors that opened from the tower's lowest level into the plaza outside. Swiftly they closed those doors, barred them and massed behind them, and then I was rushing toward the great dull-glass ball at the room's center.

Inside that ball stood the great table-map upon its great block, while beside it were the six levers and speed-knob which controlled the speed and direction of Berlin. As I took the seat before them now, I gazed about me, and saw that the great ball's interior was in effect a great periscopic screen itself, one in which I could gaze in any direction through the other great ball above the static-tower's tip. And now, gazing into it around me there, I could see that in the outer night there stretched still the giant ring of the enemy cities, of which this Berlin was the heart, surrounding our own survivors and hammering them still with that deadly fire which would soon bring them crashing to earth. Far out over that mighty field of battle, its brilliant lights and blinding heat-flares stabbing the darkness, and its thunderous roar of guns shaking the air, I could gaze; while even at the same moment I heard, high above, Connell and Hilliard and their men engaged fiercely in holding the guards in the upper tower back. At the same instant came a sudden knocking, an alarmed rapping and then a battering

and crying of voices against the great tower's door from outside; as the alarm spread from the tower's upper levels!

Disregarding all these things, I grasped the controls before me, watching the scene all about the great city through the periscopic ball about me. Swiftly I jerked open the speed-knob, at the same time slamming down one of the direction-levers; and, as I did so, I saw that the whole great city of Berlin was soaring up now above the ring of cities in which it had hovered, until it was a little above their level. And then I thrust back the lever in my hand and jerked down another; as I did so the mighty air-city of Berlin, the titanic air-capital whose controls we had captured and which lay now in my hands, was driving sidewise toward Geneva, that hung beside it in the ring. Toward it we sped, driving at top speed toward it at a height a little above it, so that our colossal base was on the level of Geneva's upper towers. And with set teeth I drove Berlin onward, and in the next moment its great base had sheared right across the upper towers of Geneva, had mowed down those great towers like blades of wheat before a reaper!

Then as Berlin drove on from above it I saw Geneva wavering in mid-air behind us for a moment, and then crashing down to earth through the night! I had mowed away the great electrostatic tower whose collection of cosmic energy held it aloft, and Geneva went crashing down to earth through the darkness like some giant comet of blazing lights plunging to doom! And then, beneath my hands, Berlin was driving still onward across and over that great ring of enemy cities, shearing now in the same way across the towers of city upon city in that ring. Stockholm and Cape Town and Bucharest fell as I mowed their power-towers from them; and to them that awful spectacle of Berlin rushing upon them and sending them to doom, crashing across their great towers, must have been utterly stunning and inexplicable in that wild moment!

On—on—around the great ring I held, almost insane with wild fury and excitement in that moment of triumph, driving through the night on our captured air-city and sending city after city whirling to death. I was dimly aware that the fighting above had ceased. Connell and Hilliard and their men had wiped out the guards in the tower above, and they had rushed down to defend the electrostatic tower's doors, against which a wild battering was resounding now. Huge crowds were surging madly against the tower as they felt their great city rushing through the night and crashing in

wild destruction across their fellow-cities! But, in the wild excitement that was surging through me now, I paid no attention to all about me; for surely I was swaying such colossal forces as no man ever had swayed before.

The European and the Asiatic cities were breaking from me, in wild panic, disorganized and shattered; since there came now no commands to them from this city of Berlin that had held their commander. And as they broke into a disorganized mass, the half-hundred American cities massed in the center, who had seen the terrible havoc that Berlin, beneath my hands, was wreaking upon their enemies, were themselves rushing to the attack once more; and all their guns were thundering toward the disorganized mass of their enemies!

Up toward Berlin from that mass as we rushed forward there rose to meet us the giant air-capital of Peking, battered, scarred; its commander seeking to stay this crazy destruction its sister-capital was wreaking upon their own forces. Up it came; and for an instant it seemed that Berlin and Peking must crash together bodily, but with a last wrench of the speed-control I sent Berlin racing higher. And then, as we met Peking, crashed over it, that mighty capital's power-tower also, with its other clustered towers, was sheared from it by our great base. Peking was wavering for a moment and then went whirling down to death! Yet even as it wavered, slipped and fell, its great guns were thundering savagely upon us until it had crashed to earth far below!

Victory!

And now, down through the night upon the mass of our enemy cities, I sent Berlin slanting down toward them, at its full speed, and across them in a tremendous ramming swoop that sheared the towers from a dozen of them, even as they attempted confusedly to rise and meet the onthundering mighty city! Of that confused, disorganized and broken mass there remained of them at last hardly more than a score, still savagely belching death from their guns toward our half-hundred American cities and still sending an occasional one downward! But now as I whirled the giant mass of Berlin back toward them like a striking, gigantic bird of prey, I was aware of a

tremendous battering and clanging of metal; and at the same moment Hilliard was shouting to me from the great doors that opened from the ante-rooms into the plaza.

"They're breaking down the doors, Brant!" he was screaming above the wild thunder of battle and the clamor of giant crowds that surged against those doors outside: "Fight on, though—we'll try to hold them back!"

"Hold them a moment longer!" I yelled back to him: "A moment more—!"

For now I sent Berlin whirling downward in another terrific swoop across the mass of our enemy cities and it sheared across half their mass, as they sought by rising or sinking to avoid that deadly swoop. But a half-score were left of them now, and now the half-hundred cities of the American Federation had gathered about them and were hammering them with terrific fire. No gun sounded now on Berlin, all its crews and soldiers were rushing wildly across it toward the electrostatic tower, as the city whirled and crashed and fought and ran there against their own allied cities! Caught in the terrific fire of the cities around them, the half-score European and Asiatic cities were going down with guns thundering into annihilation. But now I was aware at the same moment of a terrific uproar there at the tower's doors and of wild shouts and clanging blows there as our men fought to hold back the madly surging crowds outside!

Gripping the levers before me for one last effort, I jerked open the speed-control to its widest; and then, as it shot above the mass of the European and Asiatic cities, only a half-dozen in number now, I whirled the mighty mass of Berlin down upon them in one last tremendous swoop from which they sought in vain to swerve. They too were hesitating for a moment and then went whirling down to death, the last of the European and Asiatic cities save for Berlin itself about me! And then, as I brought that city to a stay, with the last of its companions crashing beneath and with the American cities hanging all about us now, there was a great clang of falling metal at the tower's doors, and back through them wild crowds without were pushing our black-uniformed defenders!

Connell and Hilliard at their head, our men were being pushed back through the ante-rooms, back toward the great circular room in which I sat at the controls; and, as I gazed out through the opening in the great periscope ball about me, I saw that an instant more would see them overpowering the last

of our men, rushing in upon me to take the city's controls once more! But as I saw that, I reached forward, slammed down the lever that sent the city rushing downward! I gripped that lever and with a supreme effort tore it completely from its socket! The next moment a wilder cry came from the crowds fighting through the door and the crowds over all the city outside, as they felt that city whirling and swaying beneath them, felt it whirling down to death through the night to annihilation!

And as they uttered that tremendous cry, as the swaying city flashed downward, their struggle at the door forgotten in that awful moment of doom, I was aware subconsciously that I was staggering with Connell and Hilliard and our remaining men, up the narrow stair. Up to the great tower's second level and to its windows, beside which hung our cruisers with Macklin and a skeleton crew, holding them there beside the tower even as the great city whirled with awful speed downward! Then we jumped through those windows into the cruisers. And the next moment, just as the cruisers with ourselves inside them drove upward like light from the falling city, Berlin had crashed into the earth just beneath us, with a terrific, annihilating shock that buckled it, broke it, made of it but a great twisted mass of rended metal!

Then we were driving up toward the half-hundred remaining American air-cities that hung still high above, the giant city of New York still at their center. Up until we had soared above those cities and they lay beneath us, giant circles of brilliant light, scarred here and there with countless craters of fused metal! Their great crowds of peoples had surged, now, from their towers, out into their squares and streets. They gazed as if incredulously stunned by their deliverance, at the empty air and night about them, where so shortly before had hung the enemy that had been sending them to doom. In that moment, as we hung there high over them, in that moment of incredible surprise and dawning joy, it seemed that all the world was silent after the terrific thunder of that wild rushing battle that had riven earth and sky so short a time before. It was as though the night, and the winds about us, and the white stars overhead, were as silent with astonishment in that moment as the crowds on the cities that hung around and beneath us.

Then suddenly, from those cities, there was coming up toward us as though from a single voice a single, mighty cry!

Dawnward

At dawn of the next day our half-hundred great air-cities prepared to separate, to move back to those positions that had been theirs before danger had brought them together. Through all of that night they had hung there together, their streets crowded with weeping yet rejoicing crowds; and, now that the dawn showed them the green plains far beneath, they were beginning to depart. And we five, sitting there within the periscopic screens in the power-tower of New York, were watching them as they prepared to go.

Battered and scarred, all of them, by that titanic battle through which all had fought, many of them with great towers fused and broken, and scarred with great craters, they were crowding toward us. Washington was foremost among them, until it hung just beside our city, its great streets and plazas thronged with shouting crowds. And, in the wild shouts of those crowds, we could hear our own names, their roaring tribute. For, in their eyes, we had saved them by that last wild effort of ours. And then Washington was moving away from us, toward the southeast, speeding away and vanishing as a dark spot in the southeastern sky.

And now another was crowding beside us, and another, and still they came. Pittsburgh and Guatemala, Tacoma, Chicago and Philadelphia, Rio de Janeiro, Kansas City and Vancouver—one by one they were driving beside us, their giant bulks hanging beside New York, their mighty cheers reaching us. They moved away to north and south and east and west, to vanish as dark, dwindling spots in the skies, until New York alone of them remained hovering there high above the earth. And then, Yarnall's eyes returned to the screen beneath us, where there were revealed the great, shattered wrecks, laying half-buried in the green earth far below.

"We win," he said, slowly: "The American Federation wins; but at what cost? Two-thirds of the world's cities have crashed to annihilation and death, and a half of our own."

"It's so, Yarnall," I said, gazing down with him: "yet it was our necessity, and not our will. They attacked us without warning; attacked us with

mighty weapons which they had devised especially to annihilate us all—and we could but defend ourselves."

"I know it, Brant," he said: "We could do nothing else—but I am glad—glad, man!—only that this greatest of all wars is the last."

"Earth's Last Air War!" It was Macklin speaking, thoughtfully: "Now, the war lords of our enemies have gone, their people will join with us to end all wars, to forget all! our enmities...."

"We will," I said, as I turned toward the controls, "but we five can never forget what has happened."

And then, as the others sat silent at my words, I was opening the speed-control before me, moving over one of the great levers, and sending New York, with its great motors droning and its tube-propellers hissing, away to the east; toward the sunrise, faster and faster, rushing eastward over the green plains that were now rolling swiftly back far beneath. On all the mighty city around us, in all its streets and plazas, its great surging crowds were shouting still, a great, rejoicing clamor. But we five there at the city's controls, in the great tower, sat silent and unmoving. Gazing out into the blue cloudless heavens before us as our city rushed downward, we looked into the face of the morning sun. It was the sun rising on a world at peace.

The End

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